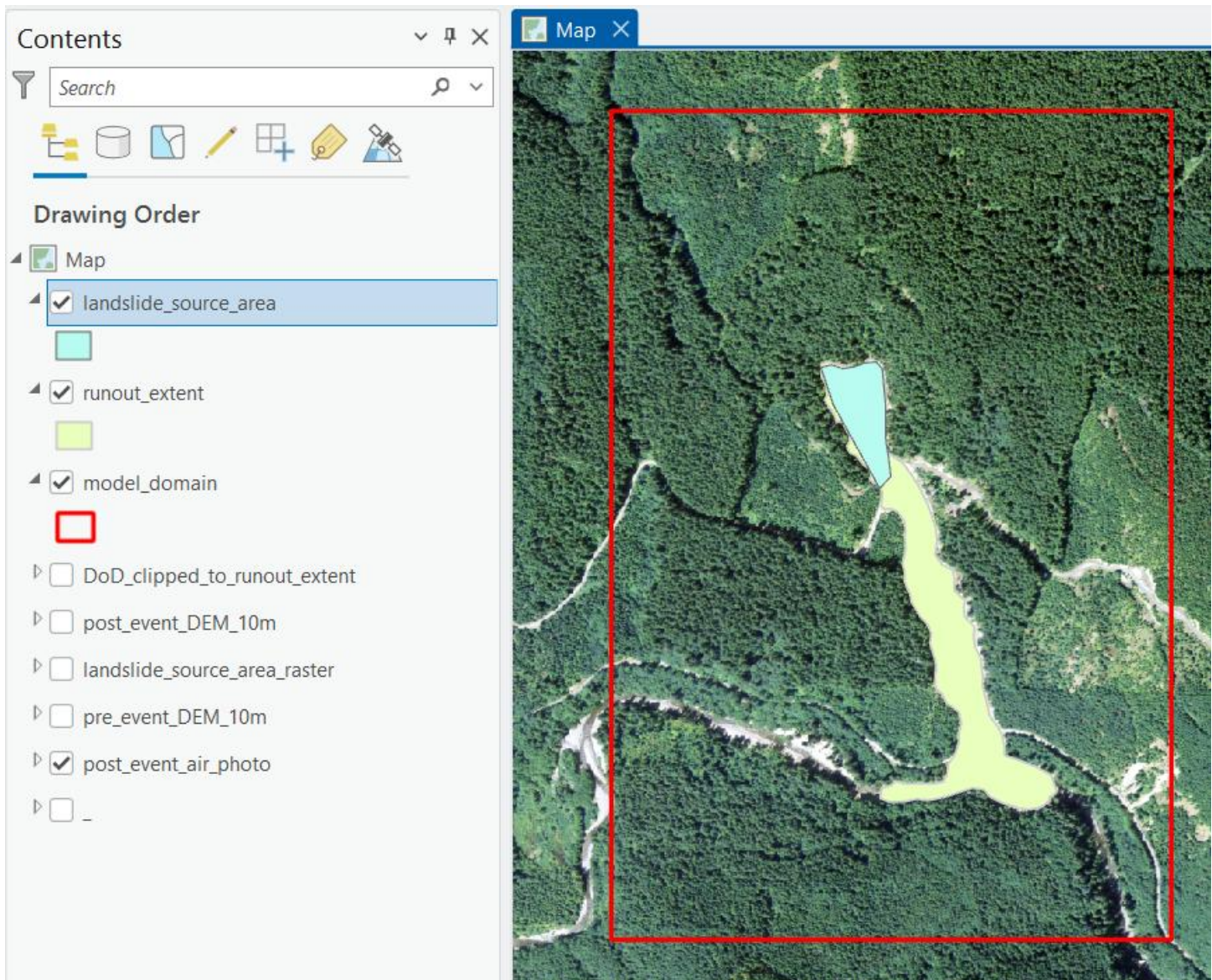


A step-by-step guide on how to prepare input data for MassWastingRunout

This guide demonstrates how to prepare inputs for MassWastingRunout, using the Cascade Mountain, 2009 site data as an example. Directions are specific to ArcPro but can be reproduced in any other GIS software/coding environment that has similar geoprocessing tools.

By the end of the guide, you will create the GIS layers shown in the Contents Pane on the left side of the figure below:



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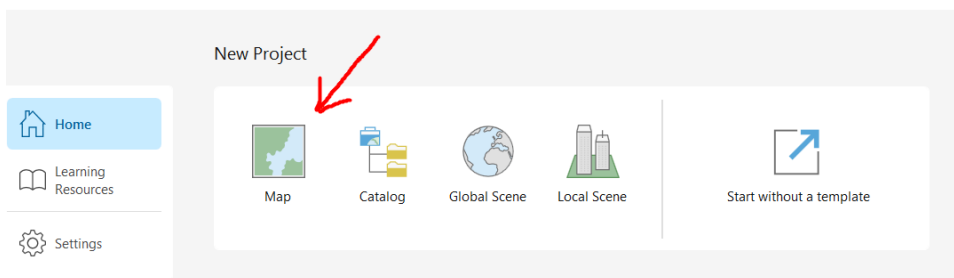
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Step 1. Setting up the ArcPro Map

1. Open ArcGIS Pro and create a new project

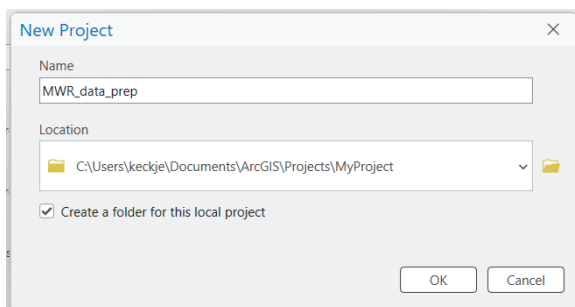


a) Select Map for the new project



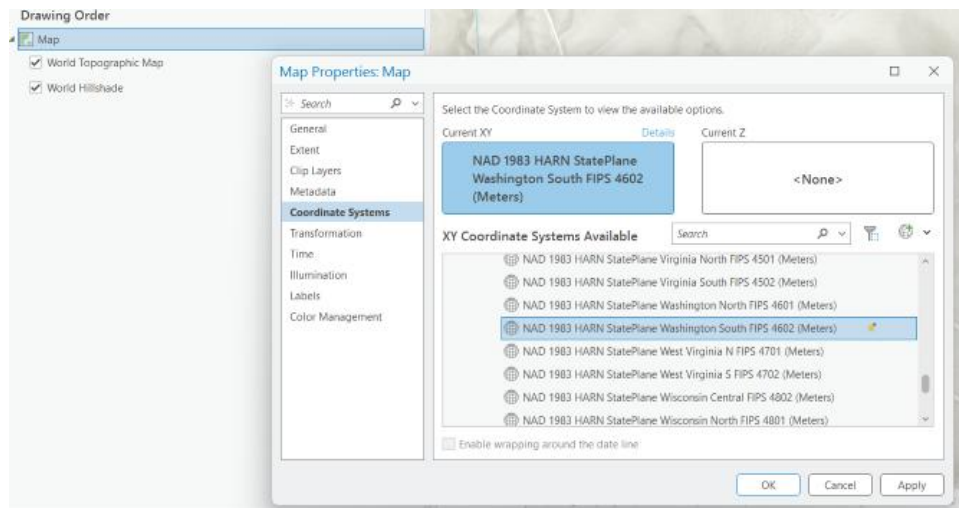
b) Pick where the project will be saved and the project name.

i) This example project is called MWR_data_prep.



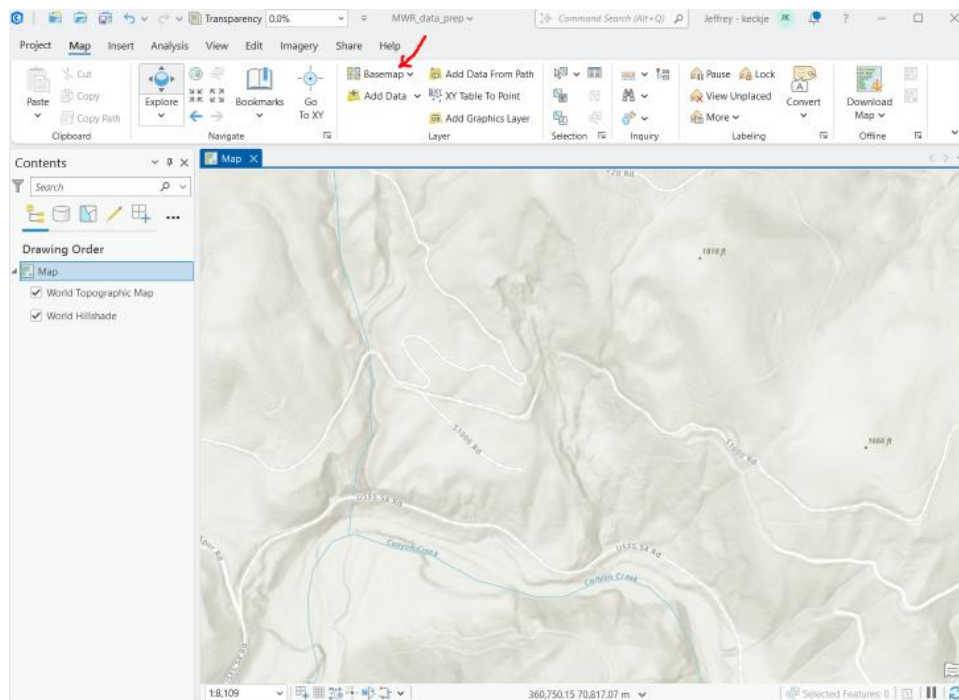
2. Set the coordinate system of the map

- The coordinate system you select here will be used to define the model grid of MWR and its inputs.
- Right click on Map in the Contents pane => select Coordinate Systems => pick a metric coordinate system. For example, for the Cascade Mountain landslide, which is in southern Washington State, I picked Nad 1983 HARN StatePlane (meters).



3. Zoom to the vicinity of your project location

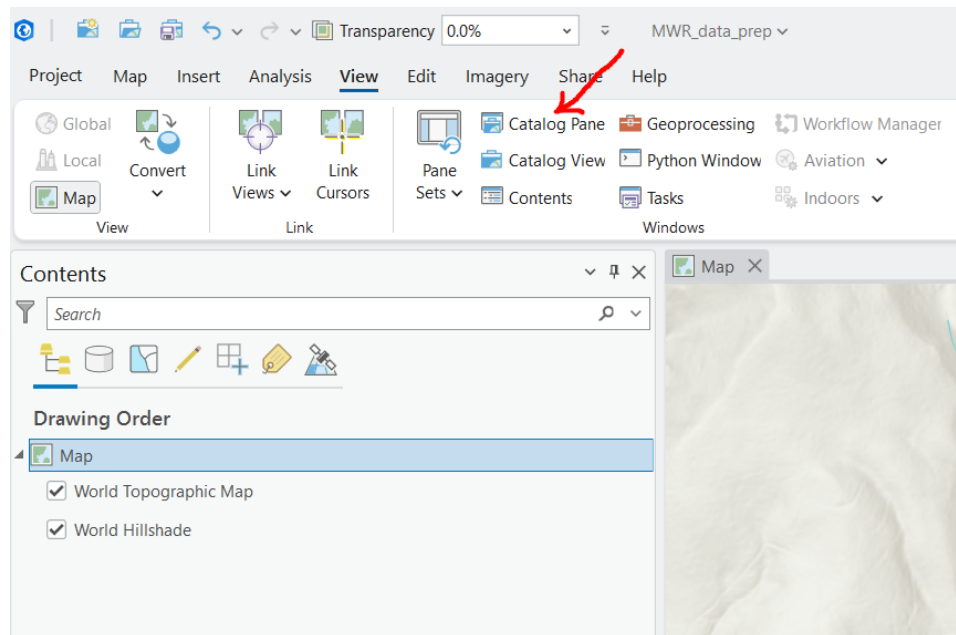
- Using the built in World Topographic Map and World Hillshade or Imagery basemaps:



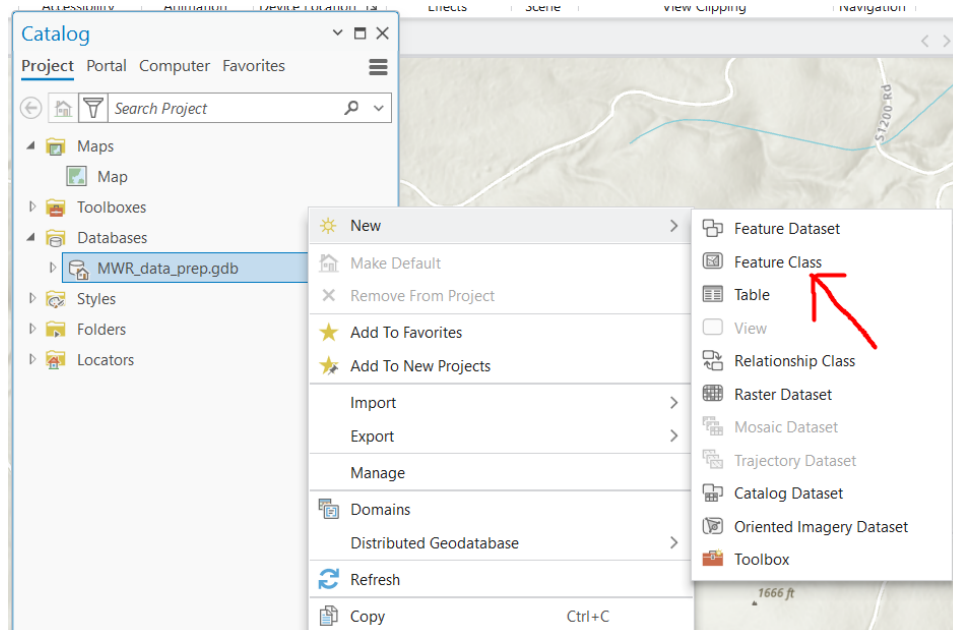
Step 2. Define the extent of the MWR model domain

1. Create a new polygon layer to represent the model domain

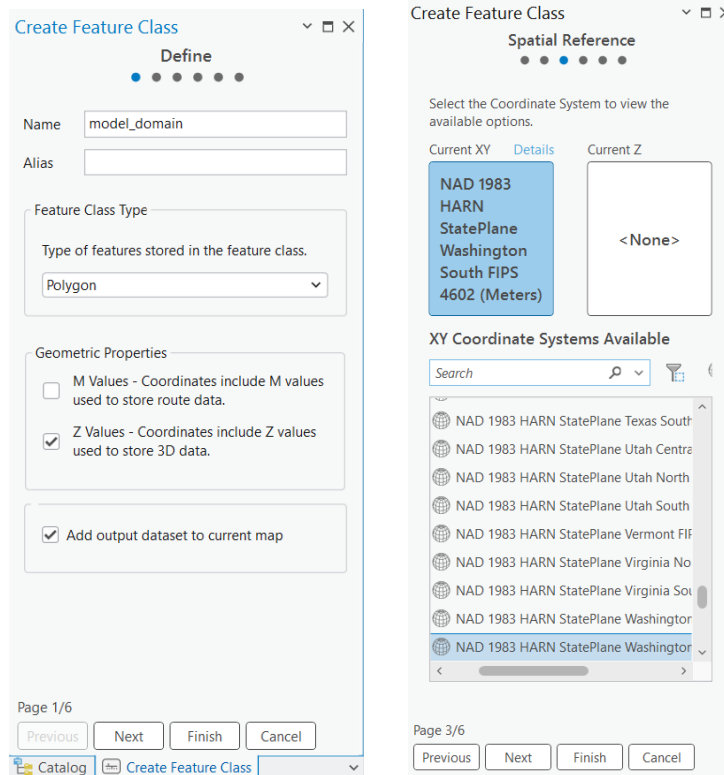
a) Turn on the Catalog Pane



b) In the Catalog Pane, navigate to the geodatabase for the project (in this case MWR_data_prep.gdb), right click the geodatabase => select “New” => select “Feature Class”. Note: “Feature Class” is ArcGIS jargon for “shapefile”.

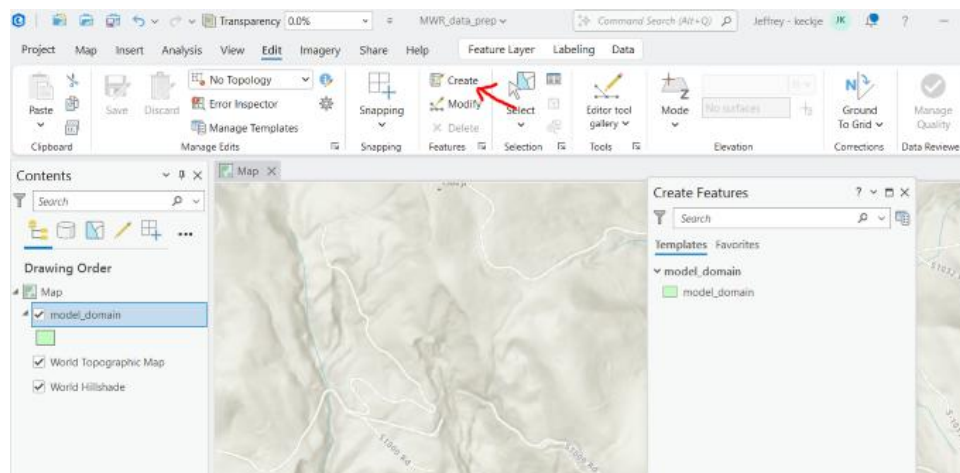


- c) Name the Feature Class “model_domain” => set the type as “Polygon”=> click next until the Spatial Reference option appears => select the same spatial reference as the project, in this case NAD 1983 Harn StatePlane Washington South (meters), and click Finish.

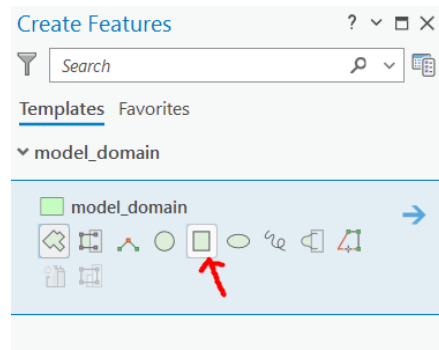


2. Draw the model_domain polygon

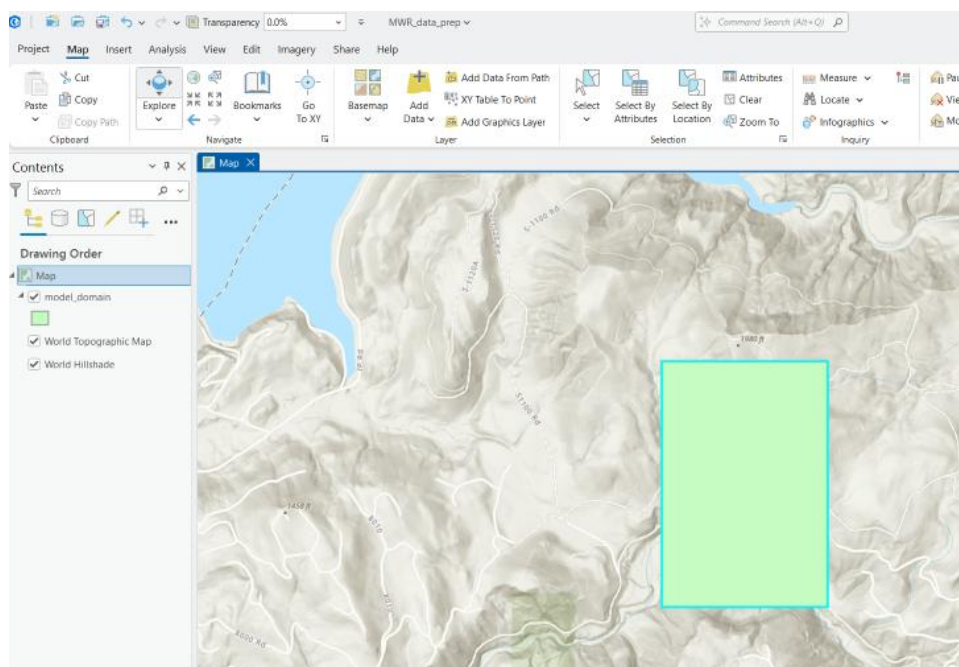
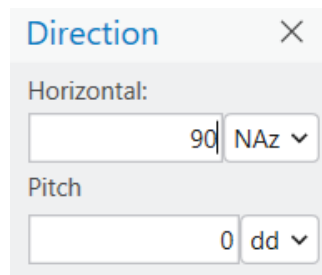
- a) Go to the Edit ribbon, click “create”, select “model_domain” from the Create Feature pane that pops up on the right.



b) Select the rectangle drawing option



c) Start drawing by left clicking the map to define one corner of the model domain, then right click to set the direction to an east-west direction as shown below. Then move the mouse to the opposite corner of the model domain and left click again to finish defining the model domain.



Step 3. Prepare and load a pre-event DEM into the map

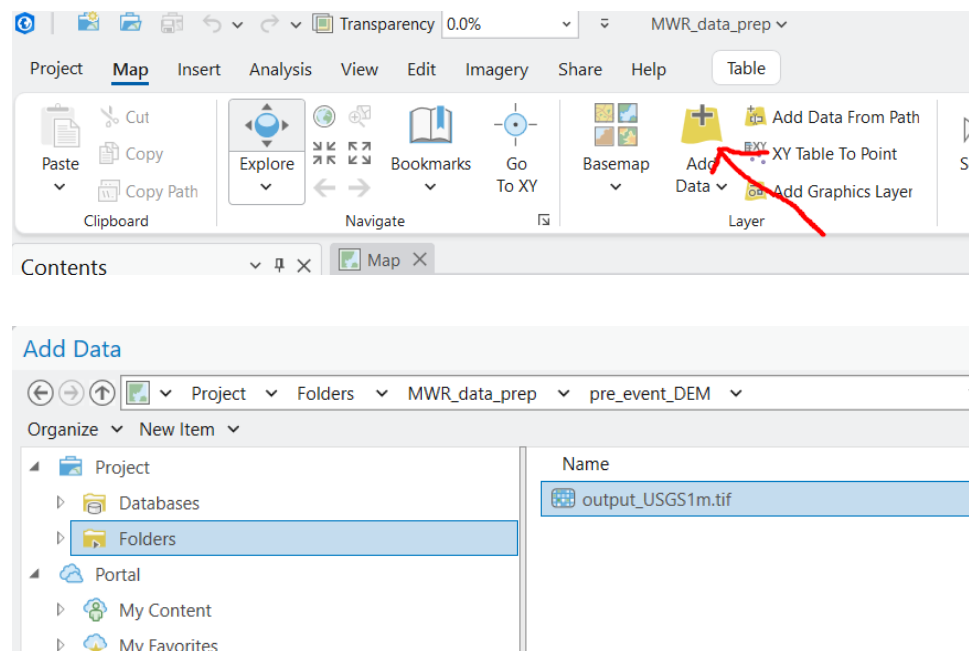
1. Prepare a pre-event Digital Elevation Model (DEM)

- a) A pre-event DEM may be available at <https://opentopography.org/>
- b) Or at a State DNR lidar portal (e.g., in Washington State: <https://lidarportal.dnr.wa.gov/>)
- c) If you are a WA State Dept. of Natural Resources employee, you can use Lidar data available at:
\\dnr\agency\app_data_gis\qdl\core\LiDAR Derivatives

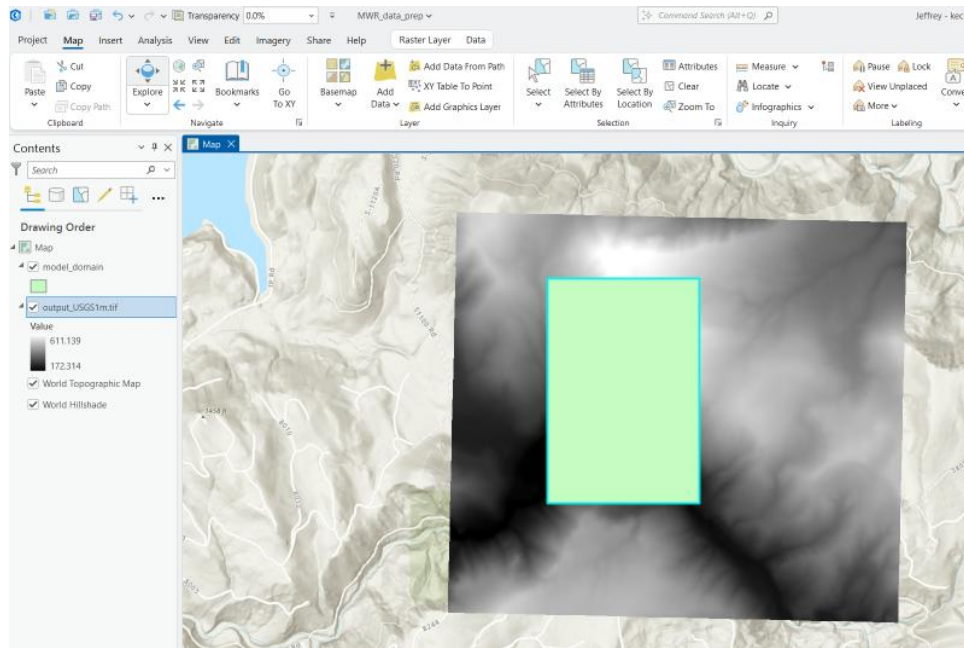
Note: Make sure the extent the DEM extends beyond the extent of the model domain

2. Add the pre-event DEM to your project

- a) Go to the Map ribbon, select “Add Data”, navigate to where you saved the pre-event DEM data, select it and hit “OK”

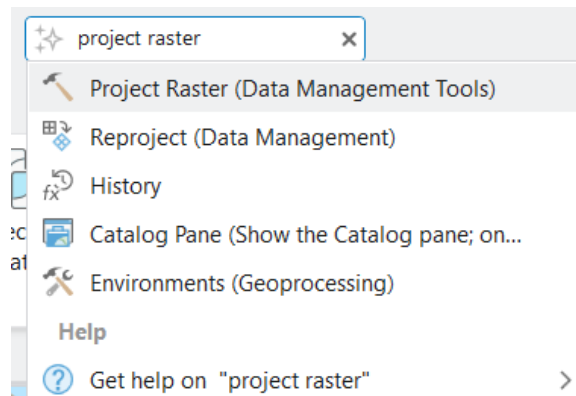


b) Confirm that the pre-event DEM extends beyond the model domain polygon



3. *Reproject, clip and resample the DEM to match the extent of the model domain*

c) First use the project raster tool (find this in the search bar) to reproject and resample the DEM to the coordinate system and grid size to be used for the MWR model setup



- d) In the project raster tool, pick the pre-event DEM you prepared, name the new file “pre_event_DEM_resampled_to_10m” => select the output coordinate system, which should be the same as your Map or the model_domain shapefile => select the geometric transformation. ArcPro may select this automatically. If the DEM will be downscaled from a fine resolution to a coarser resolution DEM, select the resampling technique => define the output cell size. In this case I am using a 10 m grid size.

The screenshot shows the 'Project Raster' tool interface within the ArcGIS Pro Geoprocessing pane. The tool is titled 'Project Raster' and has tabs for 'Parameters' and 'Environments'. The 'Parameters' tab is active. The tool's parameters are as follows:

- Input Raster:** A dropdown menu showing 'output_USGS1m.tif'.
- Output Raster Dataset:** A text box containing 'downloaded_pre_event_DEM_10m'.
- Output Coordinate System:** A dropdown menu showing 'NAD_1983_HARN_StatePlane_Washington_'.
- Geographic Transformation:** A dropdown menu showing 'WGS_1984_(ITRF00)_To_NAD_1983 + '.
- Resampling Technique:** A dropdown menu showing 'Nearest neighbor'.
- Output Cell Size:** A dropdown menu showing '10'.
- Registration Point:** Two text boxes for 'X' and 'Y', both containing '10'.

At the bottom right of the tool, there is a 'Run' button with a play icon.

- e) Next, use the “Clip Raster” tool (again access from the search bar) to clip the resampled, reprojected DEM to the model domain => here define the processing extent as the same as the model_domain shapefile but *don’t* select “Maintain Clipping Extent”. If you do select “Maintain Clipping Extent”, the grid size will be adjusted to fit the model_domain polygon and instead of a 10-m grid size, you will get an odd grid size like 8.675309 m => Do select the save location and label the clipped file “pre_event_DEM_10m” => once all settings are correct, hit “Run”.

The screenshot shows the 'Clip Raster' tool in the Geoprocessing pane. The 'Parameters' tab is active. The 'Input Raster' is set to 'downloaded_pre_event_DEM_10m'. The 'Output Extent' is set to 'model_domain'. The 'Rectangle' section shows the 'X and Y Extent' with values: Top: 71443.3094529845, Left: 361015.62853004, Right: 361809.38011755, Bottom: 70226.2236854807. The 'Extent Coordinate System' is 'NAD 1983 HARN StatePlane Washington South FIPS 4602'. The 'Output Raster Dataset' is 'pre_event_DEM_10m'. The 'Use Input Features for Clipping Geometry' checkbox is unchecked. The 'NoData Value' is '3.4e+38'. The 'Maintain Clipping Extent' checkbox is unchecked. The 'Run' button is at the bottom right.

Geoprocessing

Clip Raster

Parameters Environments

Input Raster
downloaded_pre_event_DEM_10m

Output Extent
model_domain

Rectangle

X and Y Extent

Top: 71443.3094529845
Left: 361015.62853004
Right: 361809.38011755
Bottom: 70226.2236854807

Extent Coordinate System
NAD 1983 HARN StatePlane Washington South FIPS 4602

Output Raster Dataset
pre_event_DEM_10m

☐ Use Input Features for Clipping Geometry

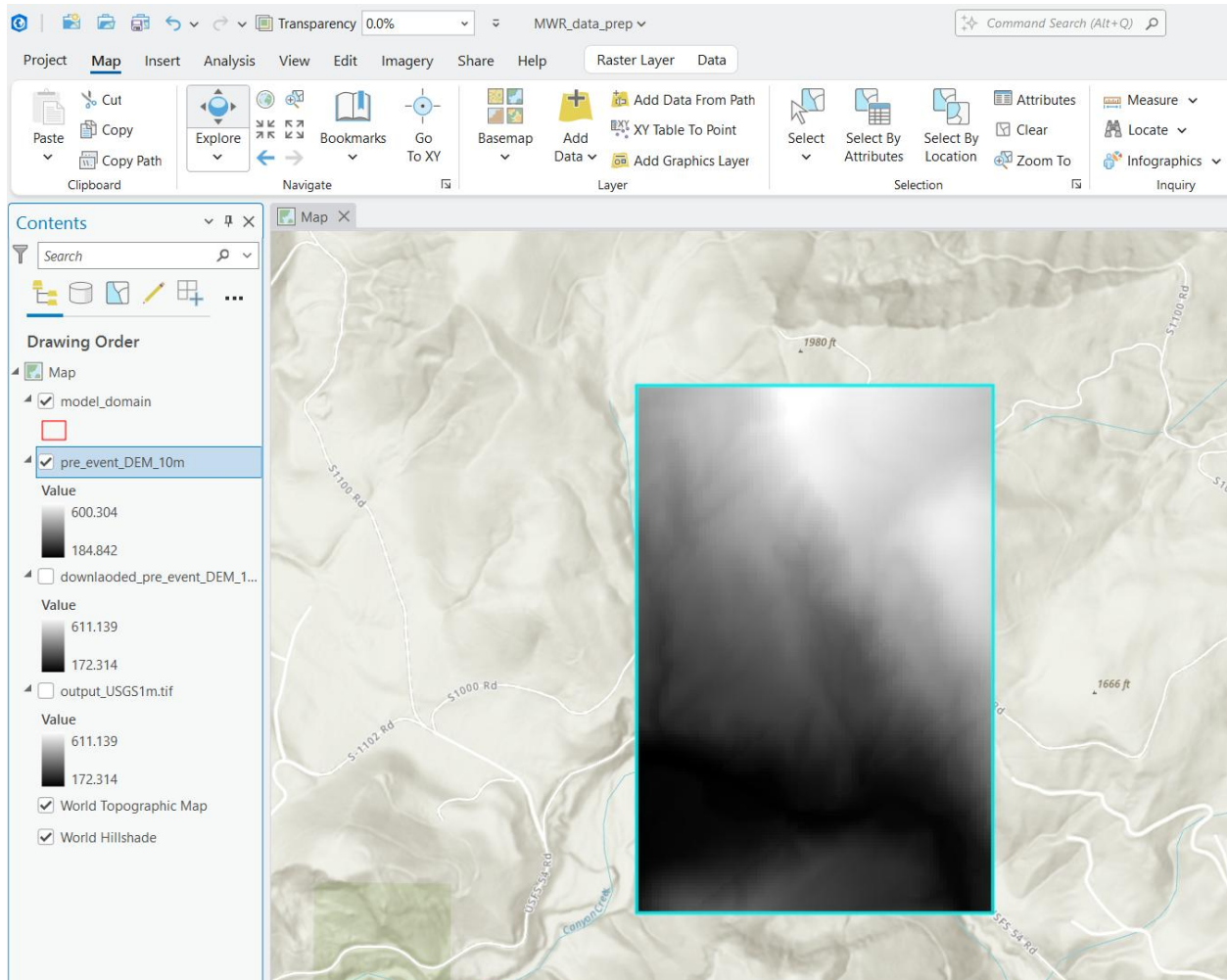
NoData Value
3.4e+38

☐ Maintain Clipping Extent

Run

- f) You should end up with a resampled DEM, oriented to the coordinate system of your map, clipped to the model domain.

Note: The pre-event DEM (its coordinate system and grid size) will be used to define all other MWR model inputs in later steps so double check that the extent and grid size seem like good fit for your landslide and terrain.



Step 4. Prepare and load an image of the post-event landslide

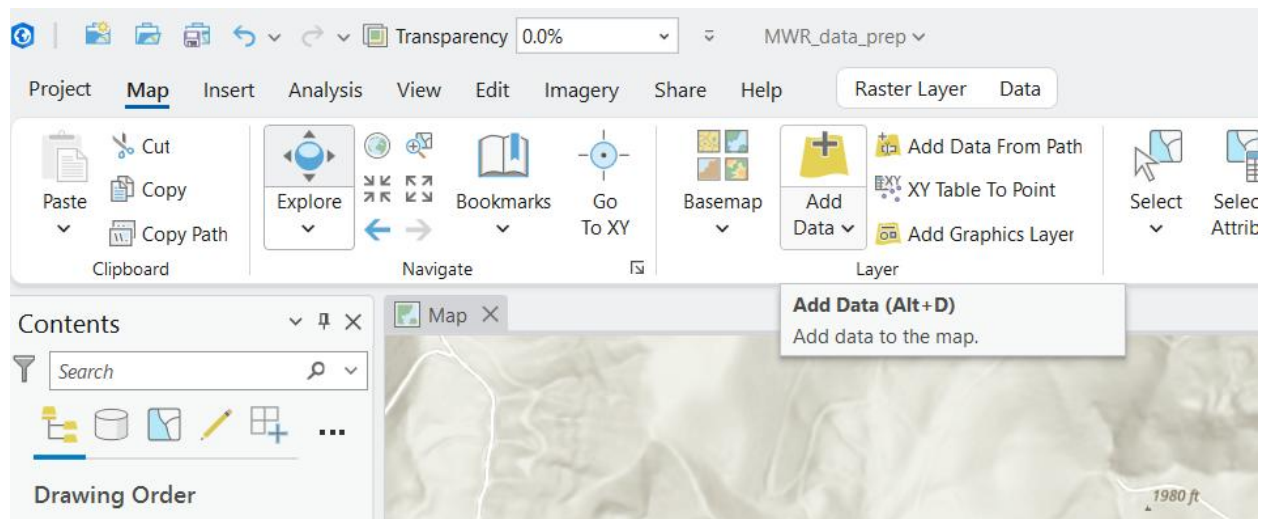
1. Prepare a post-event air photo or satellite image of the landslide

- A post-event image of the landslide may be available at: <https://earthexplorer.usgs.gov/>
- Or, if you have the means, fly a drone to get post-event images
- If available, use a State Spatial data portal (e.g., in Washington State: <https://geo.wa.gov/search?layout=grid&tags=imagery>)
- If you are a WA State Dept. of Natural Resources employee, you can find images at:
\\dnr\agency\app_data_gis\qdl\core\Aerial Imagery

Note: Make sure extent of the image includes the entire source area and runout path

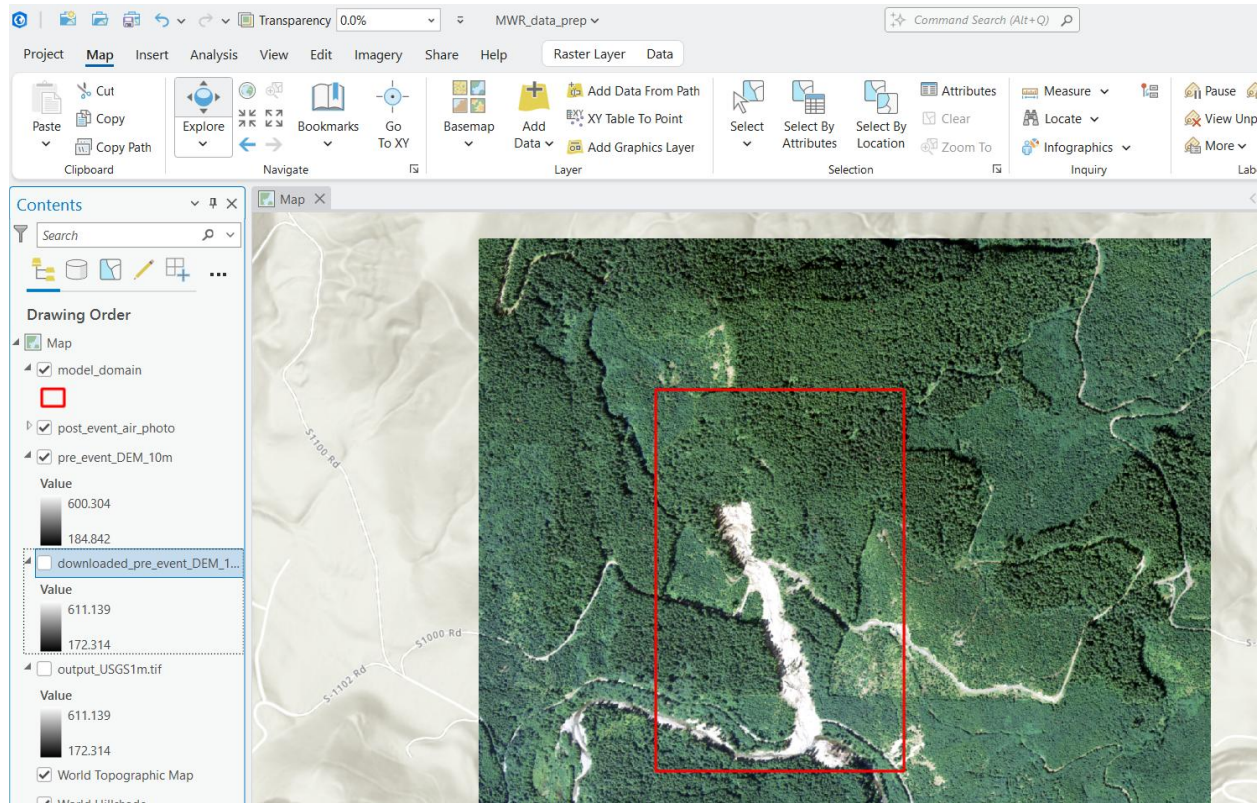
2. Add the post-event image to the ArcPro project

- In the “Map” ribbon, select the “Add Data” button and navigate to the image



- If needed, georeference the image:
 - Directions for georeferencing an image: <https://learn.arcgis.com/en/projects/georeference-imagery-in-arcgis-pro/>

- c) Once the image has been correctly georeferenced, the landslide and its runout should appear within your model domain.

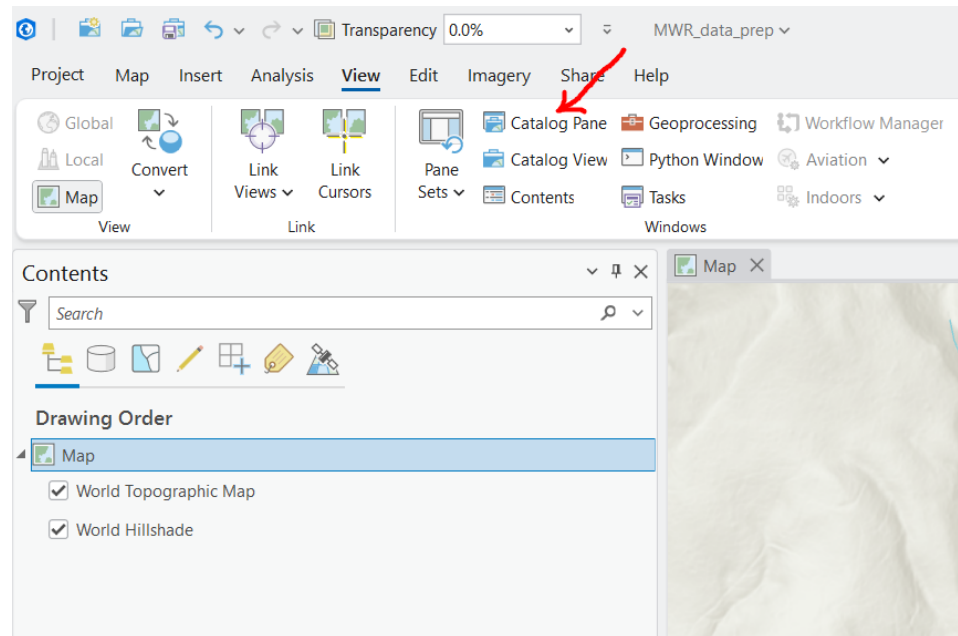


Step 5. Define the initial landslide source area

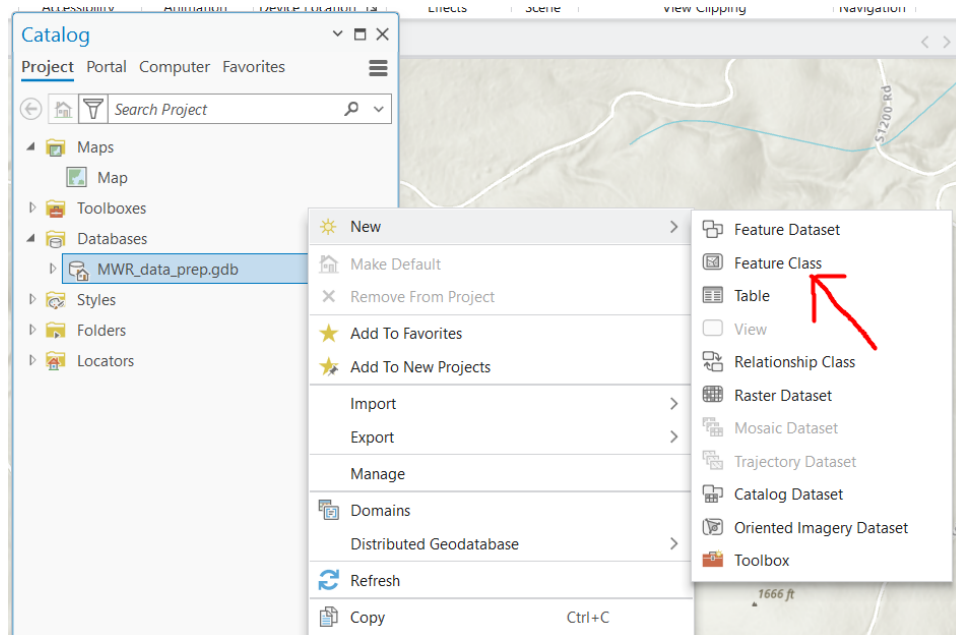
Once a georeferenced aerial image of the landslide source area and its runout is loaded into ArcPro, you will trace the outer edge of the initial landslide source area to define a landslide_source_area polygon.

1. Create a new polygon layer that will be used to define the landslide_source_area polygon

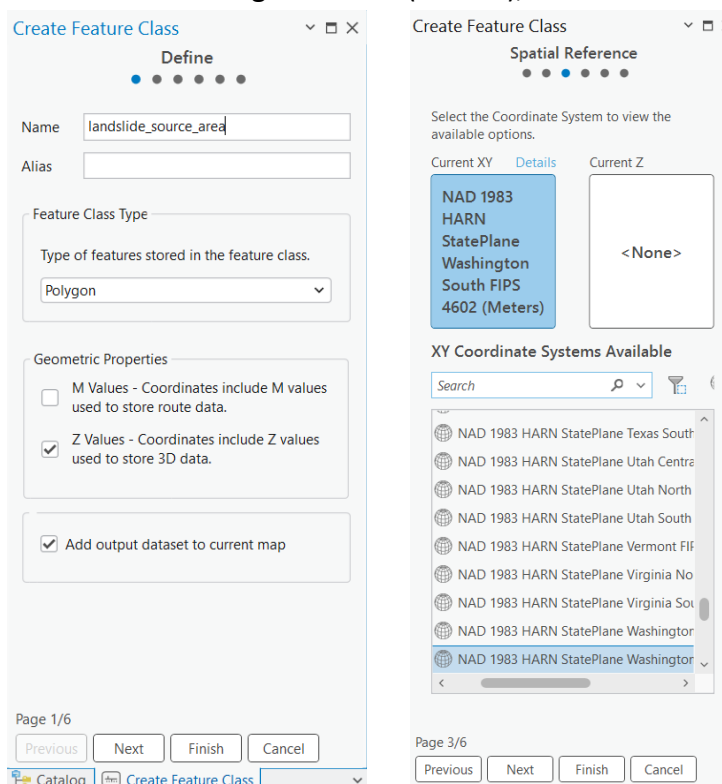
d) Turn on the Catalog Pane



e) In the Catalog Pane, navigate to the geodatabase for the project (in this case MWR_data_prep.gdb), right click the geodatabase => select "New" => select "Feature Class".

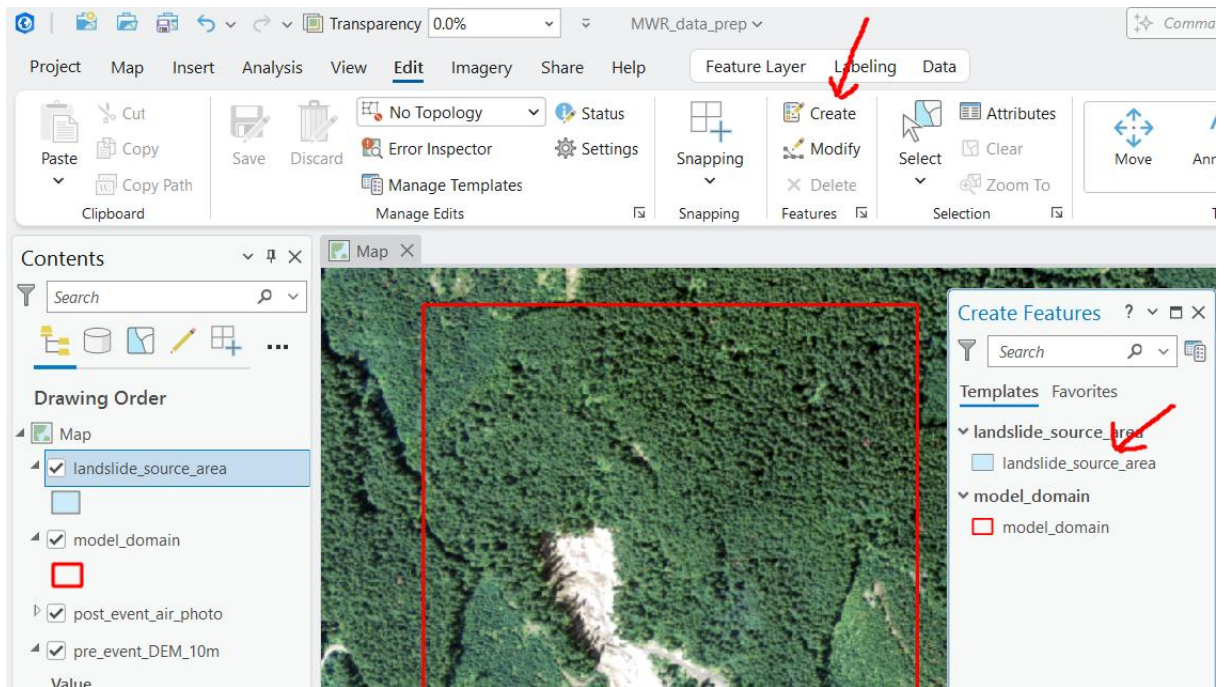


- f) Name the Feature Class “landslide_source_area” => set the type as “Polygon”=> click next until the Spatial Reference option appears => select the same spatial reference as the project, in this case NAD 1983 Harn StatePlane Washington South (meters), and click “Finish”

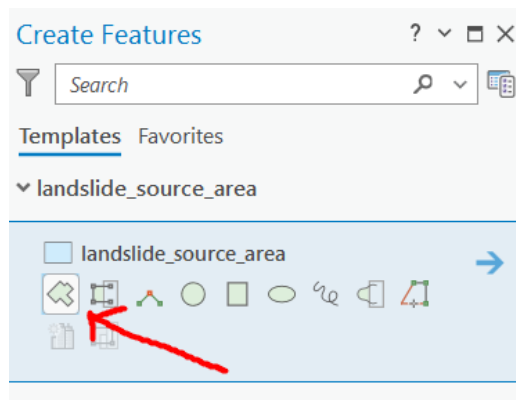


2. Draw the landslide_source_area polygon

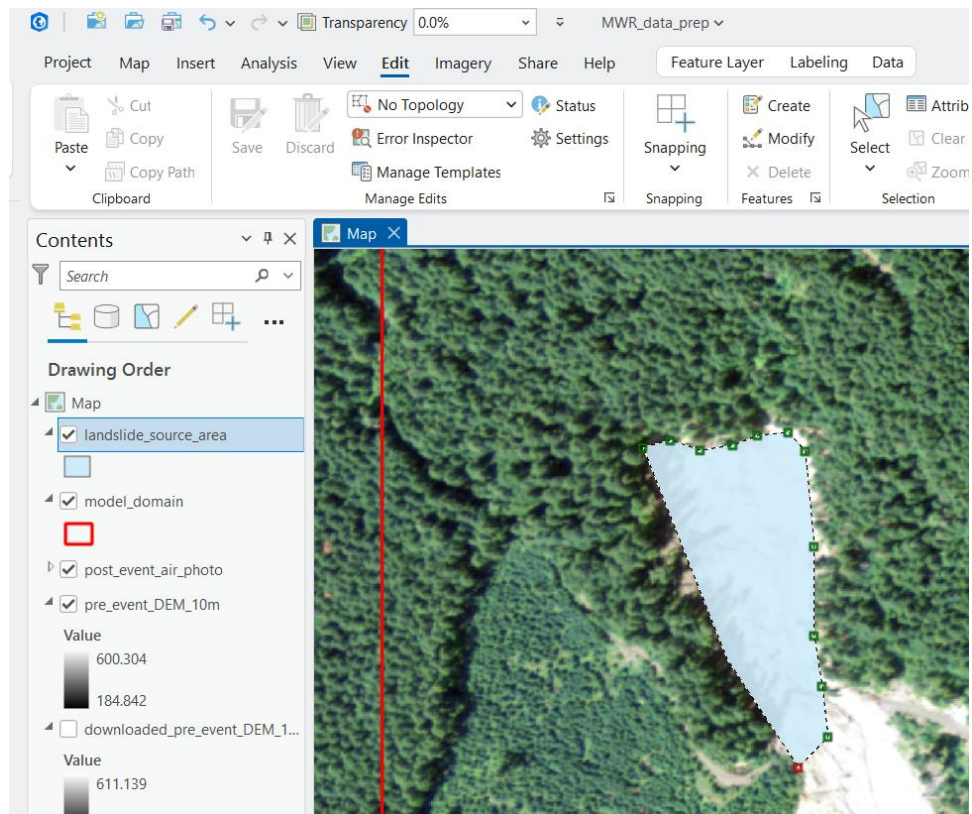
- a) Go to the Edit ribbon, click “Create”, select “landslide_source_area” from the Create Feature pane that pops up on the right.



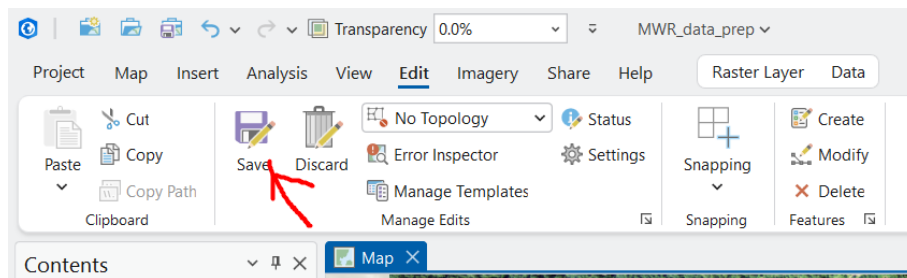
b) Select the polygon drawing option



c) Use the polygon tool to trace the outer edge of the initial landslide source material extent. Note, you may need to visit the landslide source area in the field to correctly interpret this extent as the runout and initial landslide source area can appear the same in aerial imagery.

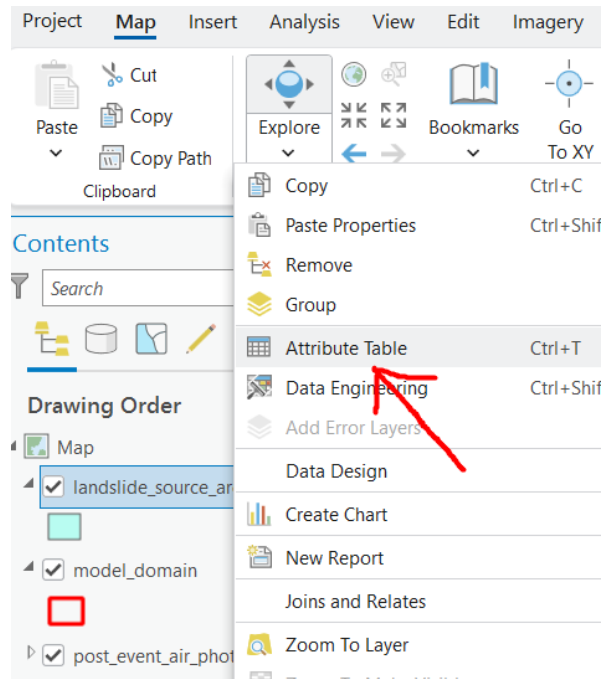


d) When you are done tracing the initial landslide source material extent, hit save

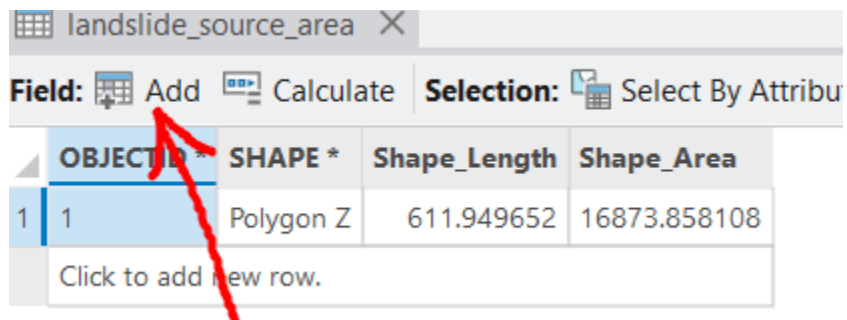


3. Add a "landslide_ID" field to the attribute table of the landslide_source_area polygon

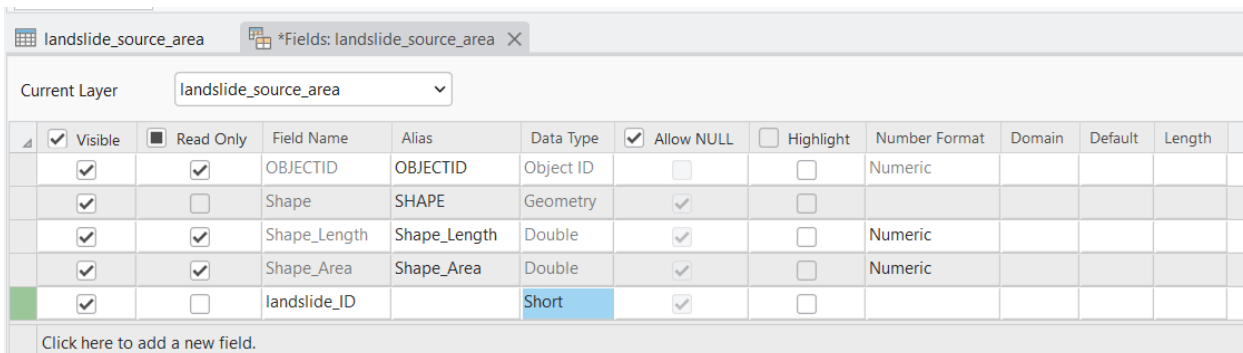
a) Right click on the landslide_source_area layer => select attribute table:



b) Click the add button above the table:



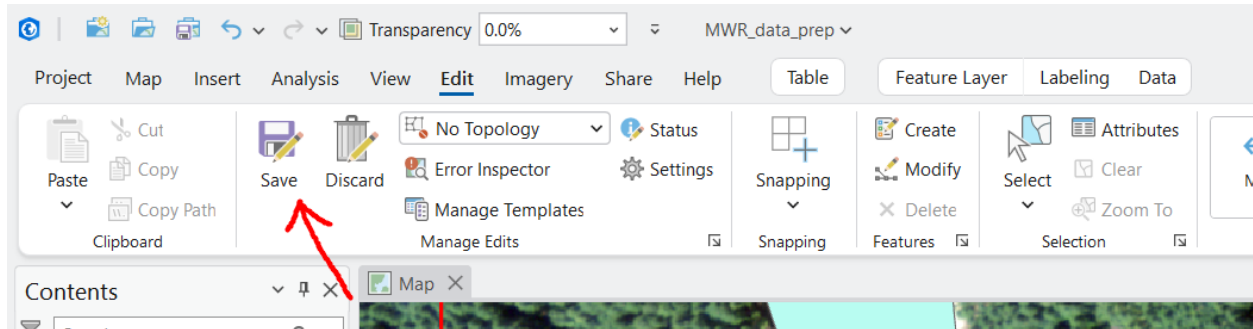
c) Click at the bottom of the table to add a new row, define the new row with a Field Name of "landslide_ID" and data type "Short" (short integer). Close the table by hitting the "x" and save when prompted.



d) The attribute table should now have a "landslide_ID" column. Double click the row for the polygon you just drew and type "1".

landslide_source_area X				
Field: Add Calculate		Selection: Select By Attributes Zoom		
OBJECTID *	SHAPE *	Shape_Length	Shape_Area	mass_wasting_ID
1	Polygon Z	611.949652	16873.858108	1
Click to add new row.				

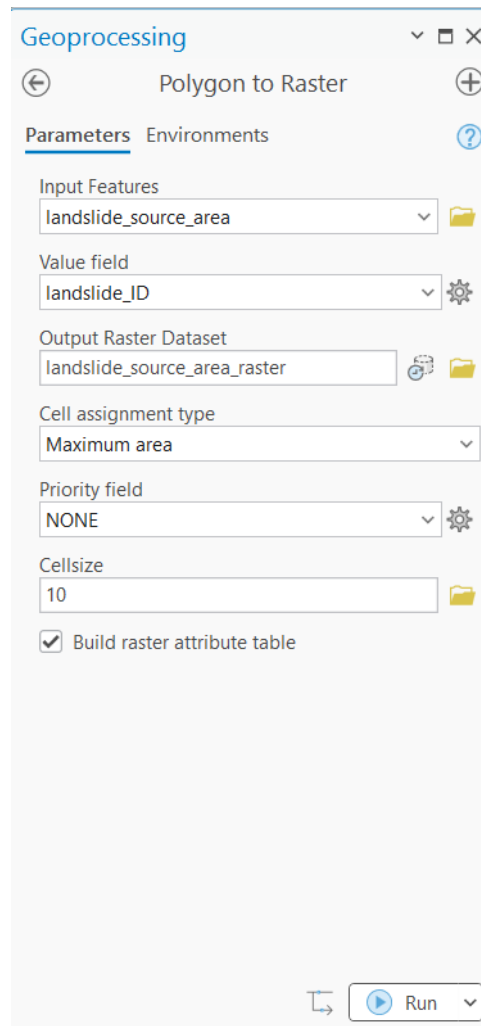
e) Then, at the top of the screen, go to the Edit ribbon, select “Save”



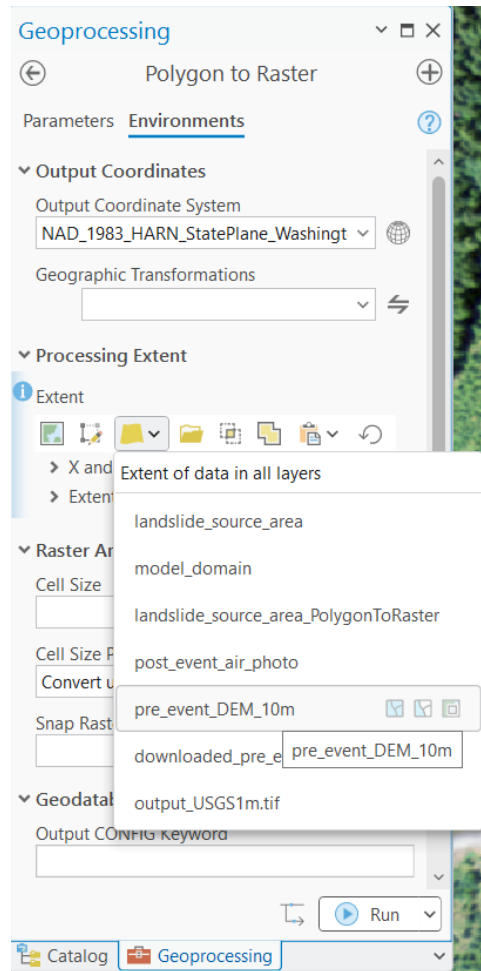
Note: If your model includes multiple landslides source areas, you can draw multiple polygons. Give each landslide polygon a unique landslide_ID value. When MWR models their runout, it will release them in sequence from the smallest ID value to the largest. If you want all landslide polygons to be released instantaneously, assign them all a value of 1.

4. Convert the *landslide_source_area* polygon to a raster dataset, with grid size and grid dimensions equal to the pre-event DEM

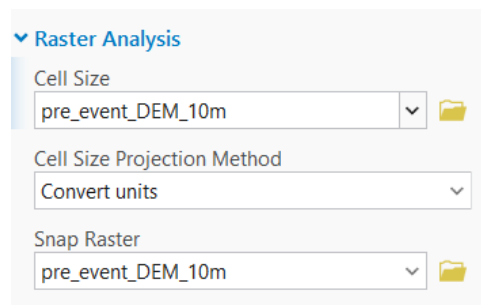
- a) Define the grid values of the *landslide_source_area* raster as the *landslide_ID* value
- i) Open the “Polygon to Raster” tool (search for Polygon to Raster tool in search bar). For input features => select the *landslide_source_area*, Value field => select *landslide_ID*. Choose where the raster version will be saved and label the raster version “*landslide_source_area_raster*”.



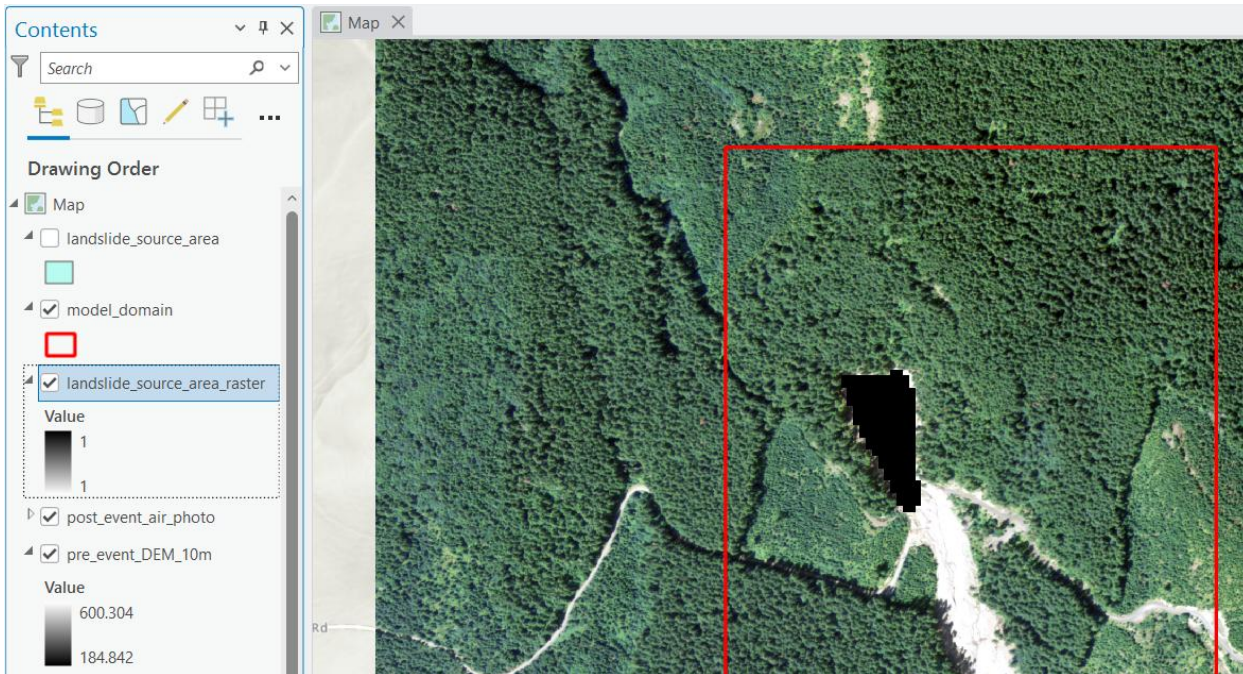
- b) Set grid size and dimensions to match the pre_event_DEM_10 layer
- i) At the top of the Polygon to Raster tool, click “Environments”. Set the output coordinate system to the pre-event DEM (pre_event_DEM_10m) layer. Set the extent to the same as the pre-event DEM layer.



- ii) Scroll down to “Raster Analysis” section, set the Cell size to the pre-event DEM and snap raster to the pre-event DEM.

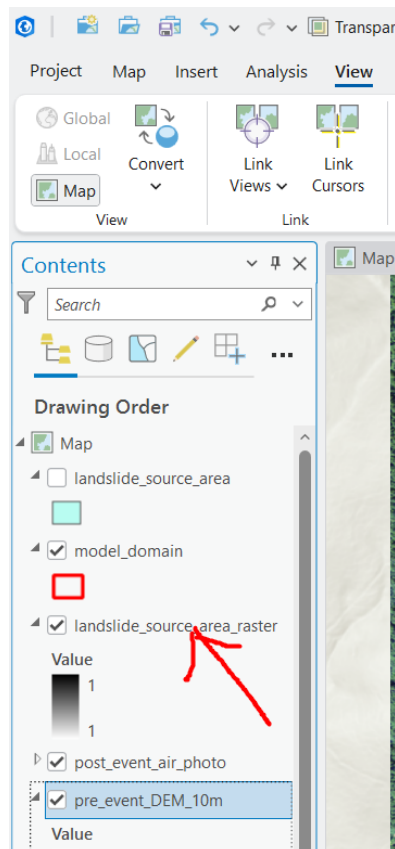


iii) Leave the Geodatabase section blank and hit run. At the Olympic Mountains, 2009 slide, I get the following:

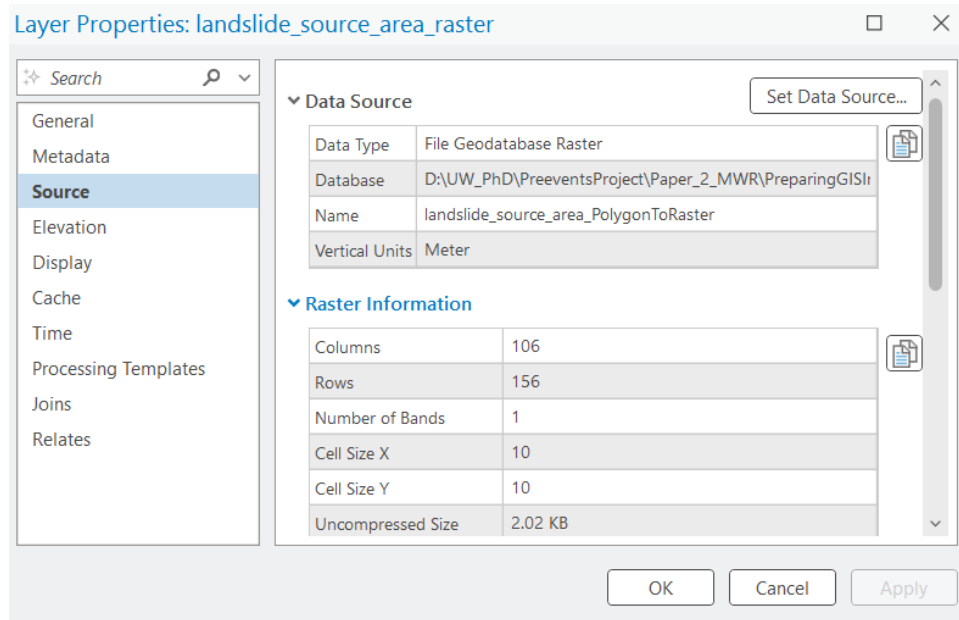


c) Check that the grid size and dimensions are the same as the pre-event DEM

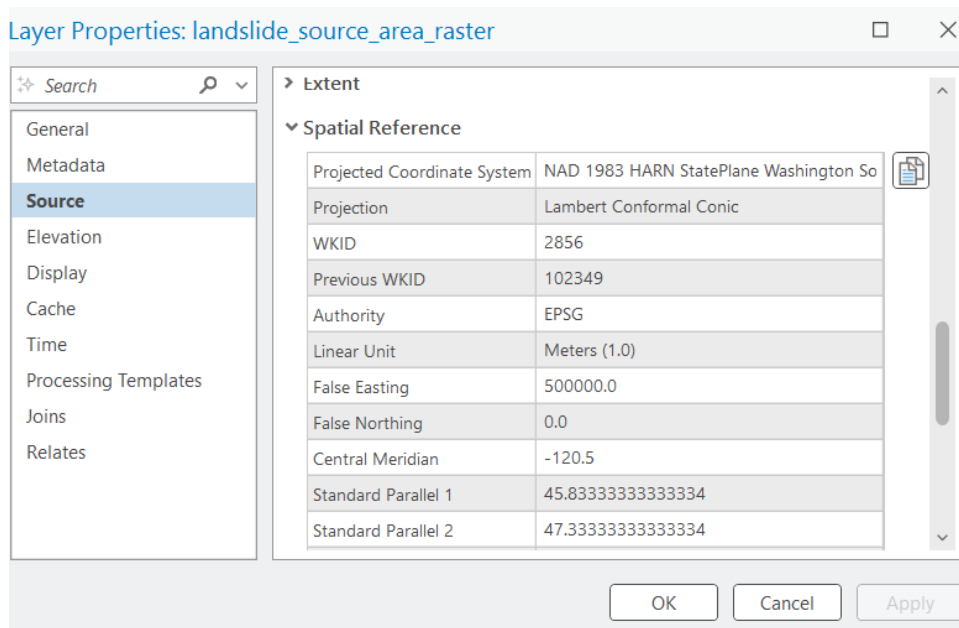
i) In the contents pane, right click on landslide_source_area_raster:



i) Select “source” and look at the Raster Information



ii) Scroll down and check the “Spatial Reference”



iii) Compare to the raster information and spatial reference of the pre-event DEM. If everything matches, then you have done everything correctly and you can move on.

Step 6. Creating the DEM of Difference (DoD)

Note: A DoD is only needed if you intend use the MWR Calibrator utility to calibrate MWR

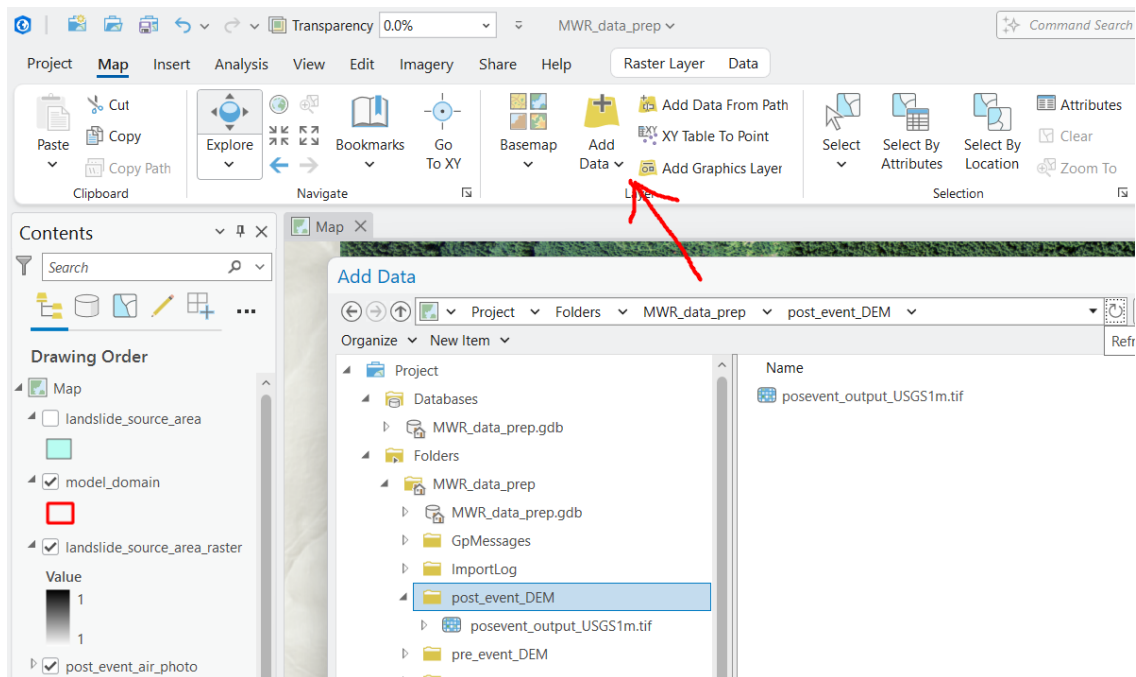
Option 1. Create the DoD from pre-event and post-event DEMs

1. Prepare a post-event and Digital Elevation Model (DEM)

- A post-event DEM may be available at <https://opentopography.org/>
- Or, if available, use a State DNR lidar portal (e.g., in Washington State: <https://lidarportal.dnr.wa.gov/>)
- If you are a WA State employee, use Lidar data available in the ArcPro quick data loader
- Use a drone to create a post-event Digital Surface Model

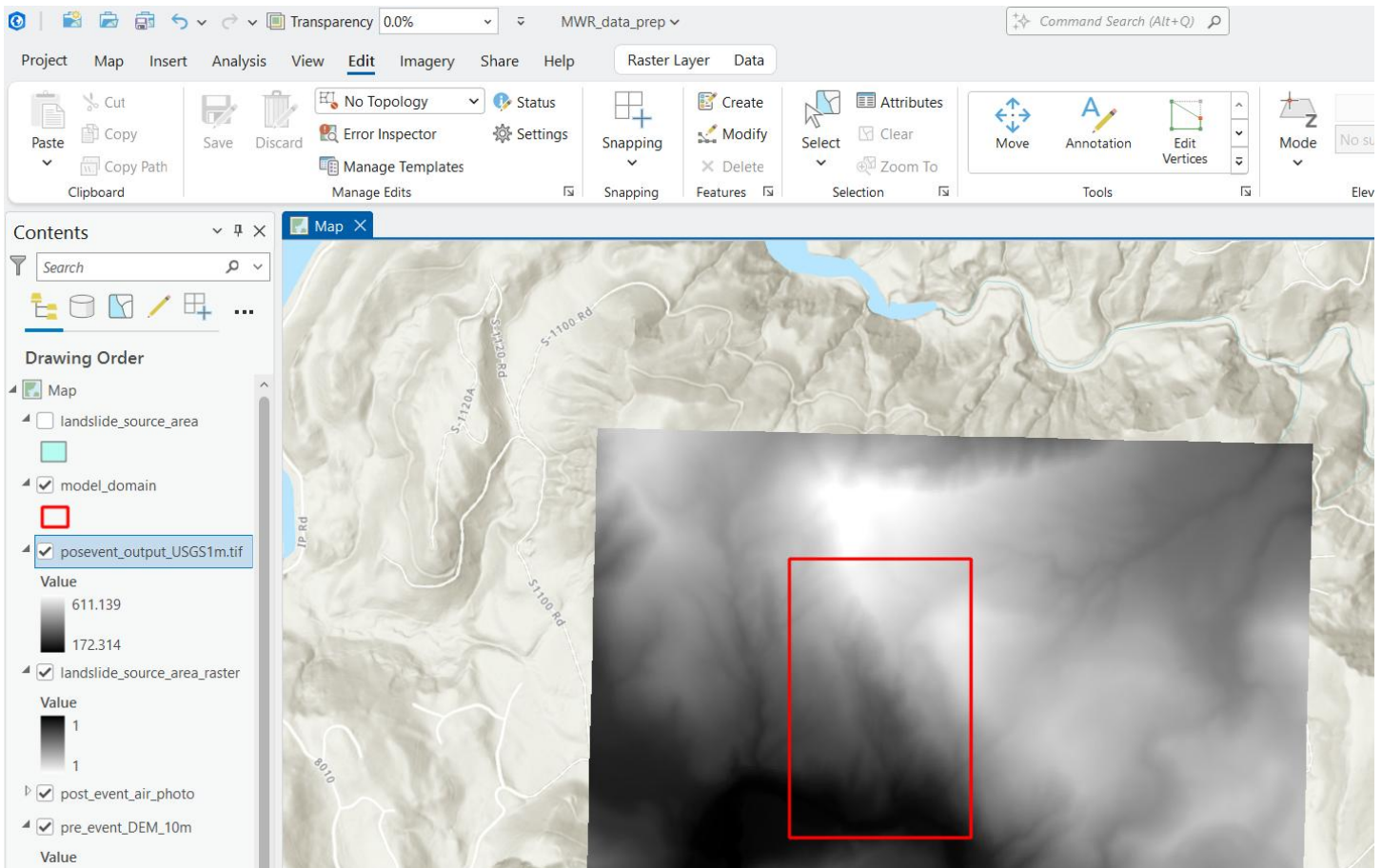
2. Add the post-event DEM (or DSM) to your project, reproject and clip it to the pre-event DEM

- Go to the Map ribbon, select Add Data, navigate to where you saved the post-event DEM data, select it and hit "OK"



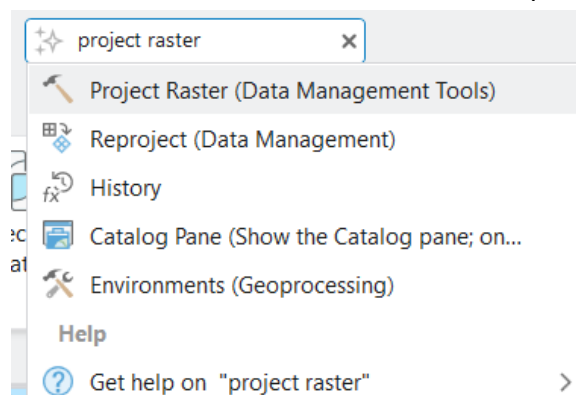
- Follow the same steps for preparing and loading a pre-event DEM section but work with the post-event DEM and label the layers you create post-event rather than pre-event.

g) Confirm that the post-event DEM extends beyond the model domain polygon



3. Reproject, clip and resample the DEM to match the extent of the pre_event_DEM_10m

- a) First use the project raster tool (find this in the search bar) to reproject and resample the DEM to the coordinate system and grid size to be used for the MWR model setup



- b) In the project raster tool, pick the post-event DEM you downloaded or created => enter a save location and name in the “Output Raster Dataset” section => select the output coordinate system to be the same as the pre-event DEM => select the geometric transformation. ArcPro may select this automatically => set the grid size to the same size as the pre-event DEM (in this case 10 m). Hit run.

The screenshot shows the 'Project Raster' tool in the Geoprocessing pane. The 'Parameters' tab is active. The 'Input Raster' is set to 'post_event_output_USGS1m.tif'. The 'Output Raster Dataset' is 'downloaded_post_event_DEM_10m'. The 'Output Coordinate System' is 'NAD_1983_HARN_StatePlane_Washington_'. The 'Geographic Transformation' is 'WGS_1984_(ITRF00)_To_NAD_1983 +'. The 'Resampling Technique' is 'Nearest neighbor'. The 'Output Cell Size' is 'pre_event_DEM_10m'. The 'X' and 'Y' cell size values are both 10. The 'Registration Point' X and Y values are empty. The 'Run' button is at the bottom right.

Geoprocessing

Project Raster

Parameters Environments

Input Raster
post_event_output_USGS1m.tif

Output Raster Dataset
downloaded_post_event_DEM_10m

Output Coordinate System
NAD_1983_HARN_StatePlane_Washington_

Geographic Transformation
WGS_1984_(ITRF00)_To_NAD_1983 +

Resampling Technique
Nearest neighbor

Output Cell Size
pre_event_DEM_10m

X 10 Y 10

Registration Point
X Y

Run

- c) Next, use the “Clip Raster” tool to clip the resampled, reprojected post-event DEM. Here, you will clip using the pre-event DEM layer. Define the processing extent as the pre_event_DEM_10m. Give the new file a name like “post_event_DEM_10m”. Don’t hit run yet.

The screenshot shows the 'Clip Raster' tool in the ArcGIS Geoprocessing environment. The tool is titled 'Clip Raster' and has two tabs: 'Parameters' and 'Environments'. The 'Parameters' tab is active. The 'Input Raster' is set to 'downloaded_post_event_DEM_10m'. The 'Output Extent' is set to 'pre_event_DEM_10m'. The 'Rectangle' section is expanded, showing the 'X and Y Extent' with the following values: Top: 71554.8778192382, Left: 360802.346630408, Right: 361862.346630408, and Bottom: 69994.8778192382. The 'Extent Coordinate System' is set to 'NAD 1983 HARN StatePlane Washington South FIPS 4602'. The 'Output Raster Dataset' is set to 'VR_data_prep.gdb\post_event_DEM_10m'. The 'NoData Value' is set to '3.4e+38'. The 'Maintain Clipping Extent' checkbox is unchecked. At the bottom right, there is a 'Run' button.

Geoprocessing

Clip Raster

Parameters Environments

Input Raster
downloaded_post_event_DEM_10m

Output Extent
pre_event_DEM_10m

Rectangle

▼ X and Y Extent

Top 71554.8778192382

Left 360802.346630408

Right 361862.346630408

Bottom 69994.8778192382

▼ Extent Coordinate System
NAD 1983 HARN StatePlane Washington South
FIPS 4602

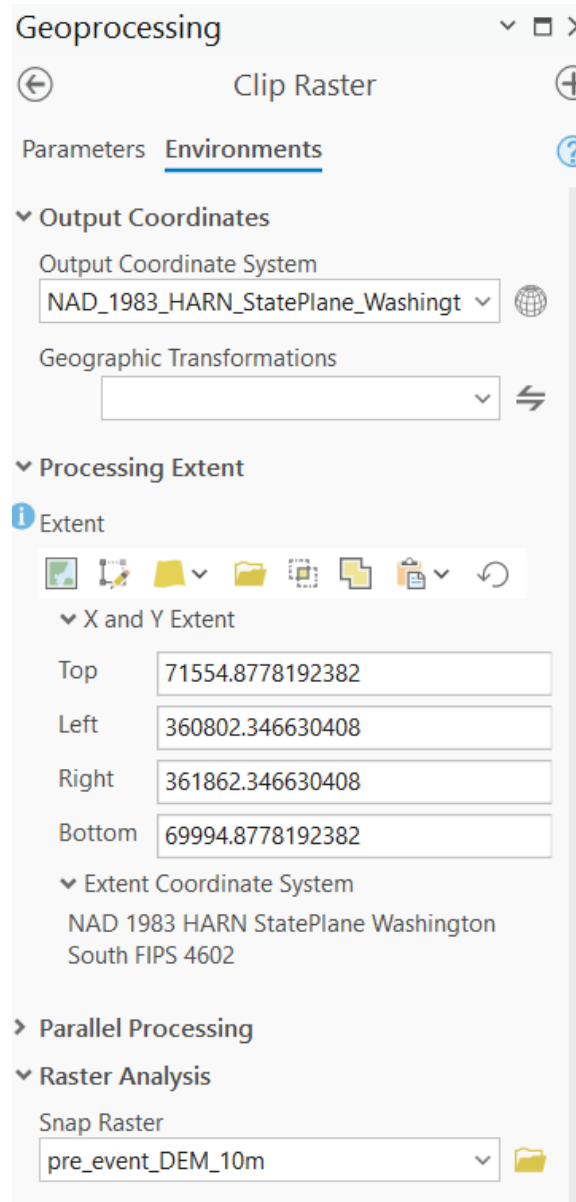
Output Raster Dataset
VR_data_prep.gdb\post_event_DEM_10m

NoData Value
3.4e+38

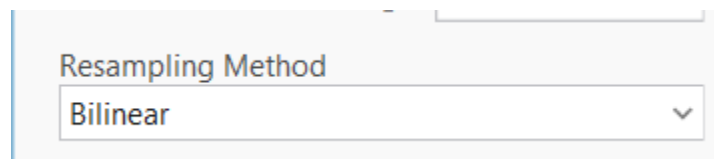
☐ Maintain Clipping Extent

Run

- d) Next, go to the Environments tab of the Clip Raster tool => set the Output Coordinate System Processing Extent and Snap Raster to the pre-event DEM.



- e) Select the resampling method. Here I use Bilinear. Now hit run.

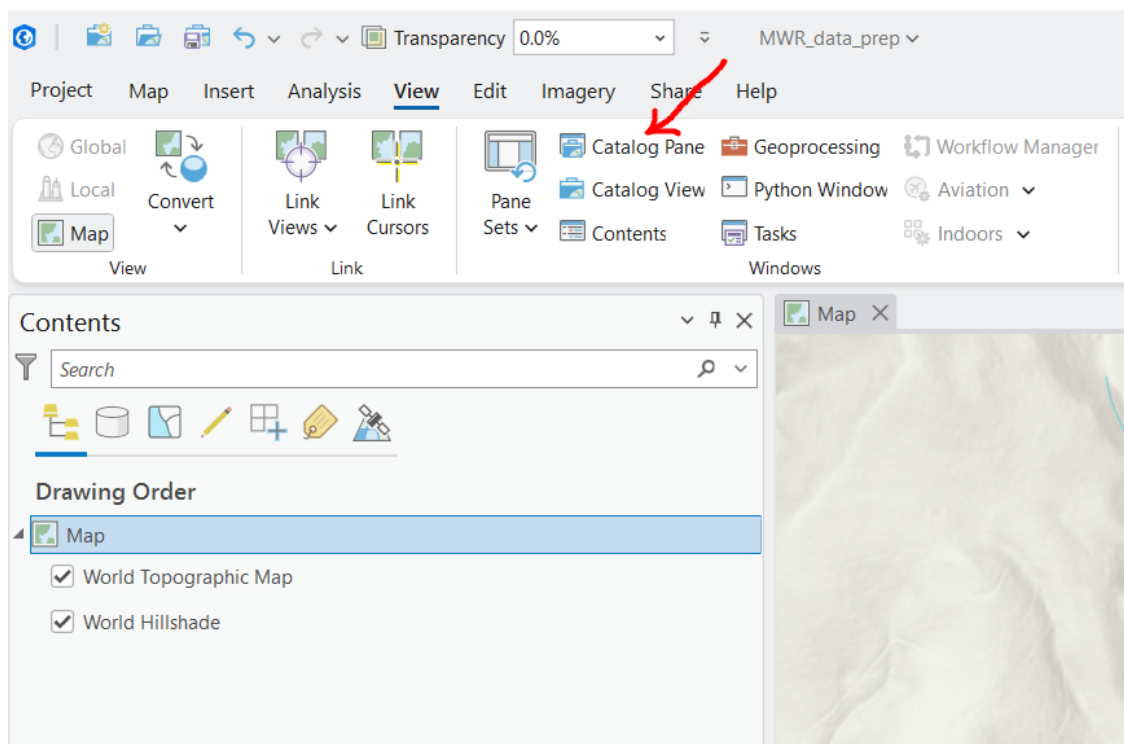


- d) Finally, check that the grid of the post_event_DEM_10m is the same as the pre_event_DEM_10m by right clicking on post_event_DEM_10m in the Contents pane and comparing the number of columns, rows and spatial reference.

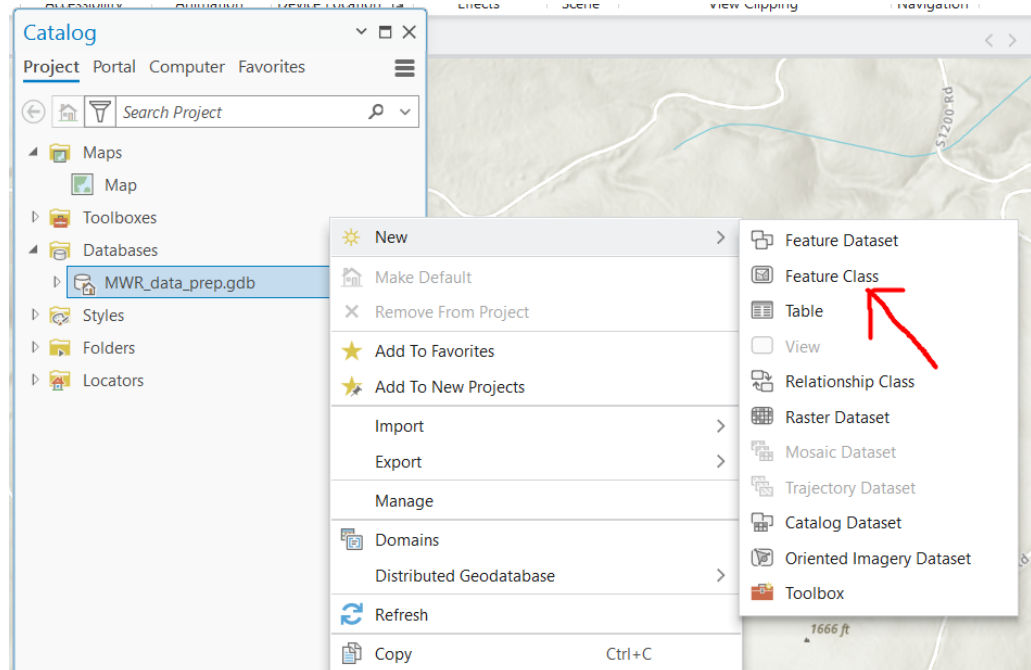
4. Define the runout path, including the initial landslide source area

The purpose of this step is to create a polygon that represents the extent of the observed runout (including the source area). This polygon will be used to exclude all changes in elevation outside of the observed runout path from the observed DoD. This is needed because an observed DoD created from repeat lidar scans or other techniques often includes noise or erroneous elevation values caused by vegetation interference or problems with how the DEMs were created. If the post-event DEM is actually a DSM, the area outside of the landslide runout may represent the elevation of the forest canopy and not the ground surface. Since the MWR Calibrator utility evaluates model performance by comparing the modeled DoD to the observed DoD, cropping the observed DoD to the observed runout extent improves calibration performance.

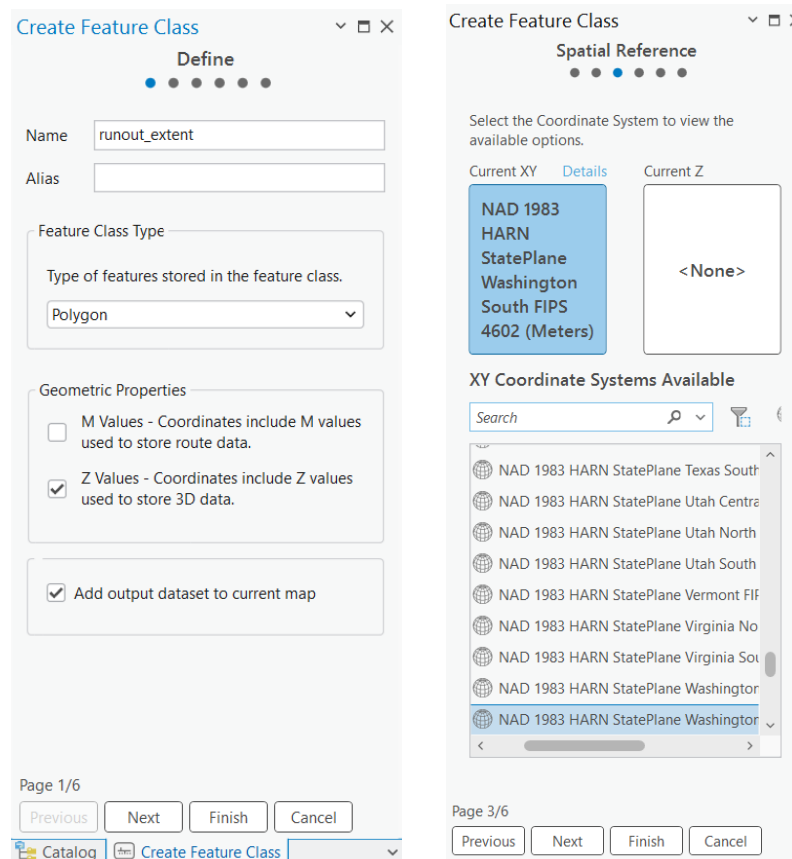
- a) Create a new polygon layer that will be used to define the landslide_source_area polygon
 - i) Turn on Catalog Pane



- ii) In the Catalog Pane, navigate to the geodatabase for the project (in this case MWR_data_prep.gdb), right click the geodatabase => select “New” => select “Feature Class”.

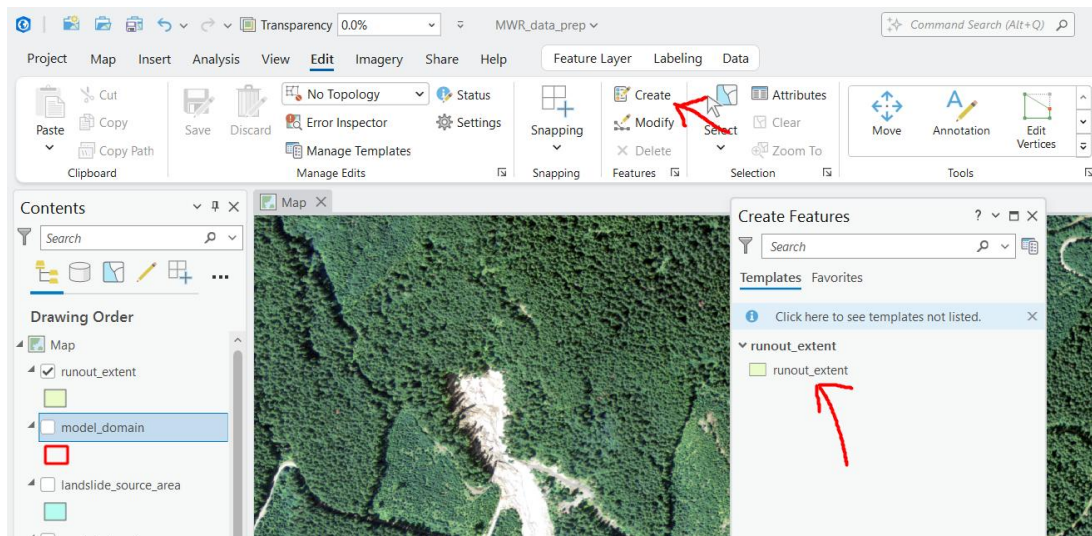


- iii) Name the Feature Class “runout_extent” => set the type as “Polygon”=> click next until the Spatial Reference option appears => Select the same spatial reference as the project, in this case NAD 1983 Harn StatePlane Washington South (meters), and click Finish

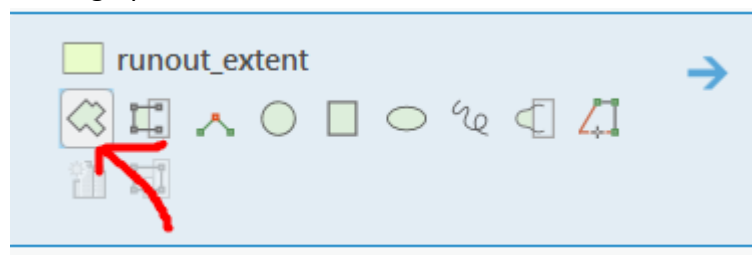


b) Draw the runout_extent polygon

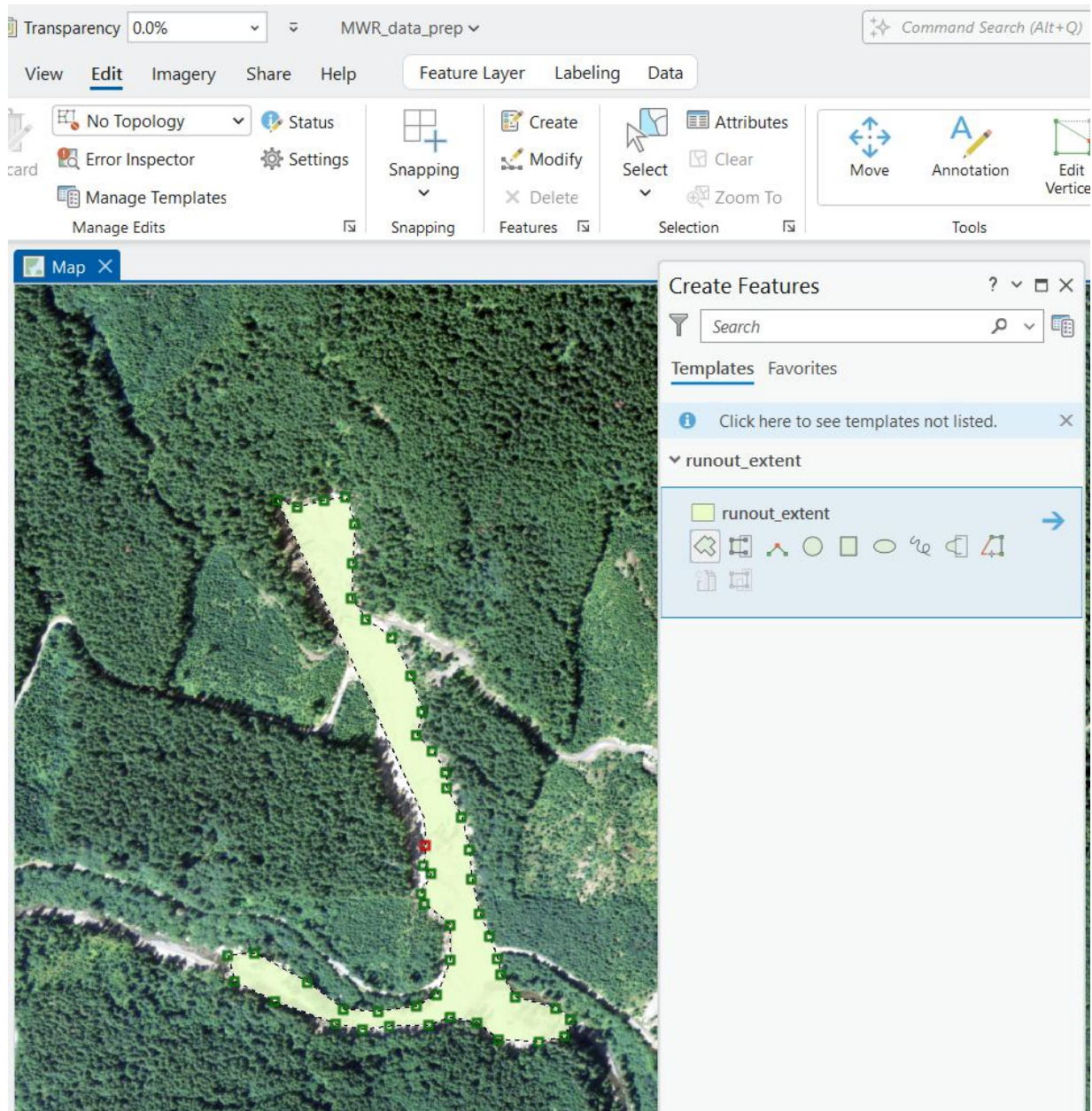
i) Go to the Edit ribbon, click “create”, select “runout_extent” from the Create Feature pane that pops up on the right.



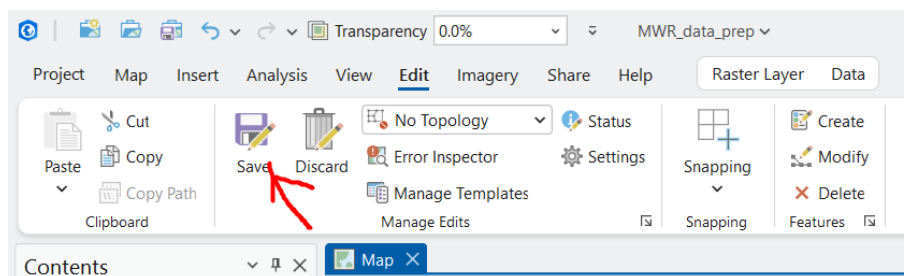
ii) Select the polygon drawing option



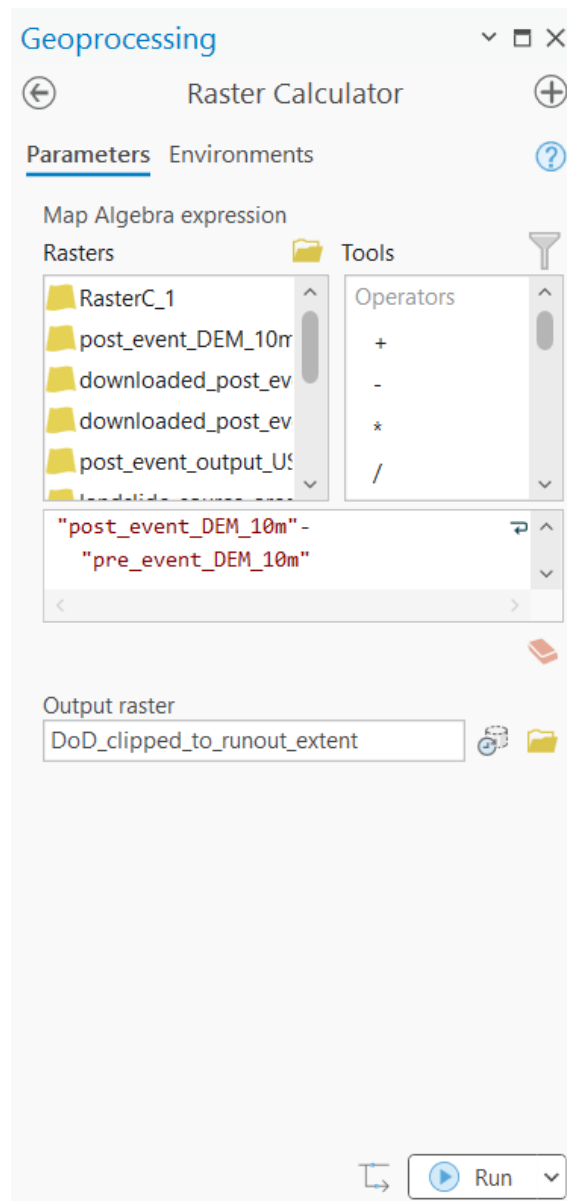
- iii) Use the polygon tool to trace the outer edge of the initial landslide source material and its runout path (see example below). You may need to visit the landslide and its runout path in the field to correctly interpret this extent.



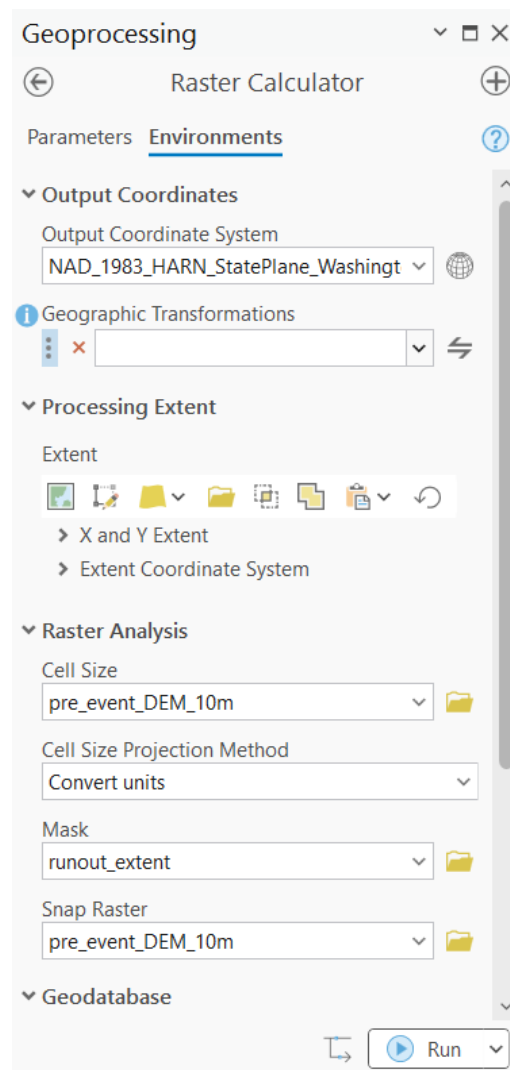
- iv) When you are done tracing the initial landslide source material extent and its runout path, hit save.



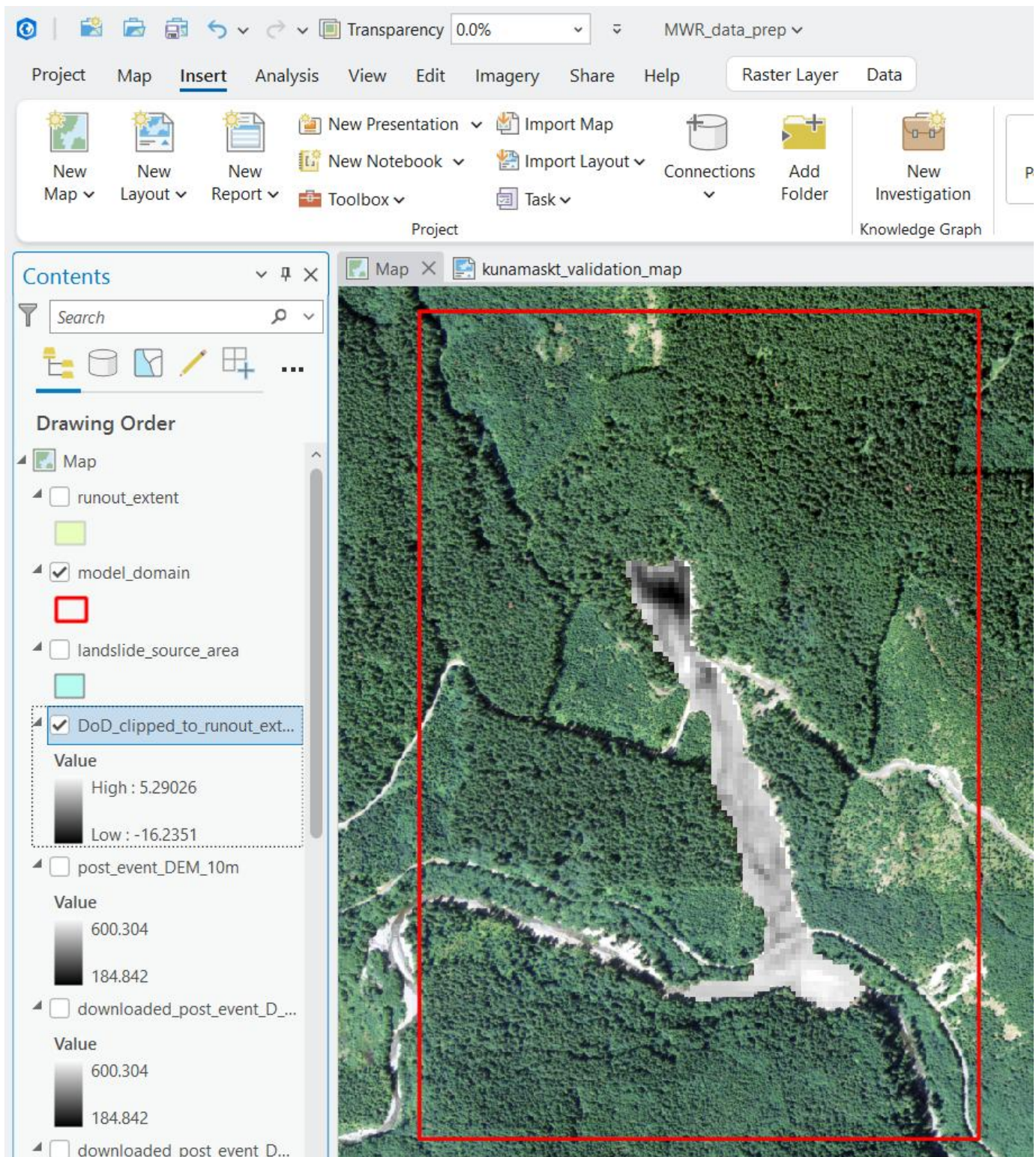
5. Subtract the pre-event DEM from the post-event DEM, clip results to the runout_extent polygon
- Open the “Raster Calculator” tool (use the search bar). In the Map Algebra expression box =>select the post_event_DEM_10m layer => select the “-” operator and => select the pre_event_DEM_10m layer.
 - Set the Output Raster save location and file name as “DoD_clipped_to_runout_extent”. Don’t hit run yet.



- c) Go to the Environments tab => set the Output Coordinate system, Cell Size and Snap Raster to the pre-event DEM. **Important** => in the Mask section, select the runout_extent shapefile.



- i) You should get a raster shapefile of the DoD, clipped to the runout extent shapefile, similar the output shown below:



6. *Check that the grid of the post_event_DEM_10m is the same as the pre_event_DEM_10m*

- a) Right click on post_event_DEM_10m in the Contents pane and compare the number of columns, rows and spatial reference.

Option 2. Create a DoD from field observations

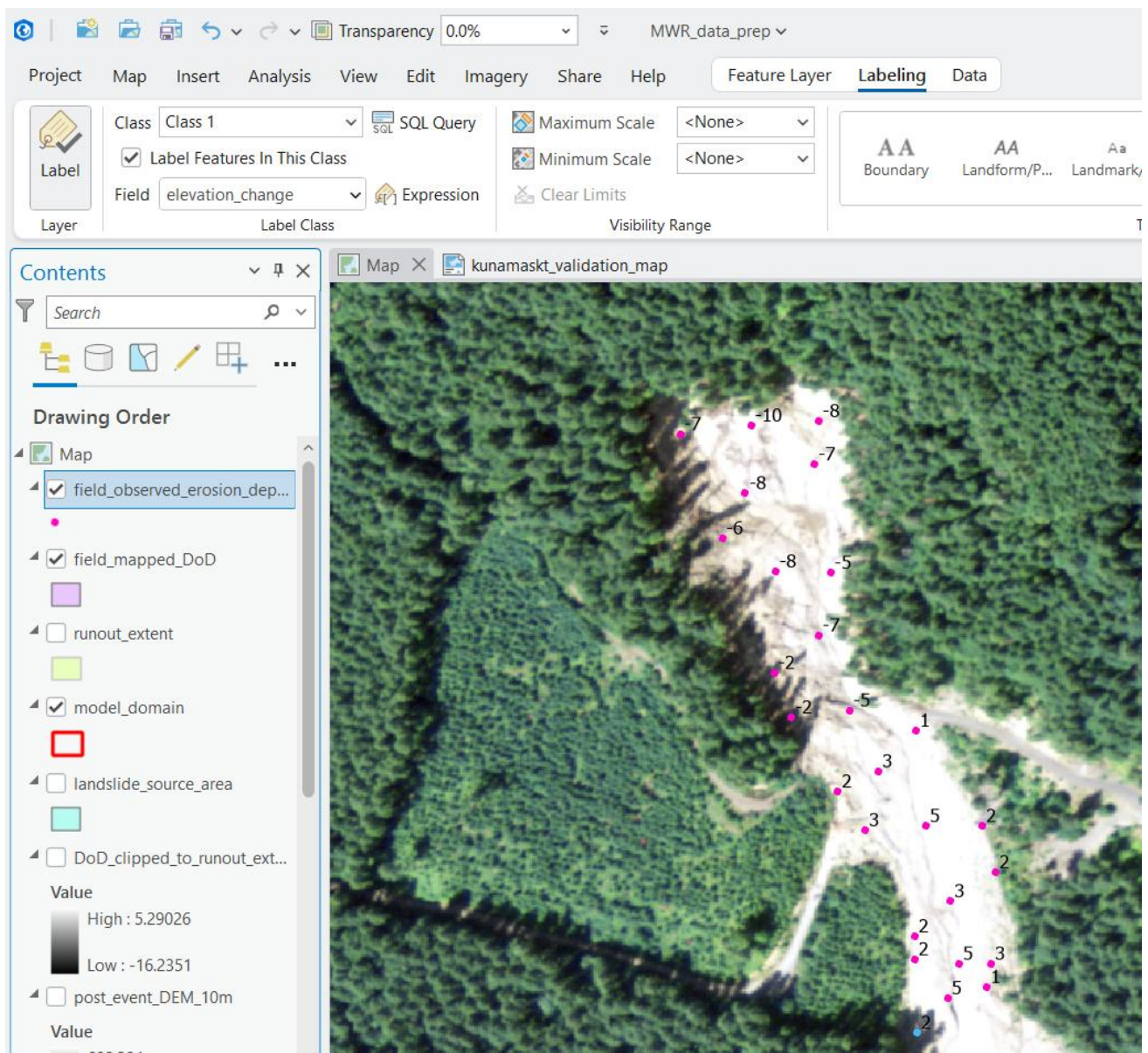
Since the final DoD map represents areas of deposition, erosion and no elevation change caused by the landslide, you can field map these areas as well and create a map in GIS. This approach involves the following steps:

1. *Hike the runout path and record erosion and deposition depths*

- a. Make sure you walk from the source area to the lowest extent of the slide or vice versa.
- b. Estimate deposition depth from the thickness of fluvially eroded terraces and burial depth of stumps or trees.
- c. Estimate erosion depth from vegetative indicators such as the remnants of in-situ tree roots or the height of the adjacent, uneroded regolith surface relative to the eroded regolith surface.
- d. Record a gps point at each measurement.
- e. As you hike, sketch zones of erosion and deposition relative to easily-recognized terrain features on a paper map.

2. Back in the office, load your gps points into the ArcPro project used to create the MWR inputs

- Call this layer “field_observed_erosion_deposition” or some other relevant name.
- Add a field to the attribute table for the observed erosion or deposition depth (see Step 5. Define the initial landslide source area for steps, but use a float data type rather than integer).
- In the attribute table, type in the erosion (- change) or deposition (+ change) for each point. Below shows example field observed depths for the Cascades Mountains, 2009 site, labeled by a change_elevation column:



3. Create a shapefile called *field_mapped_DoD*

- f. In the Catalog Pane, right click on MWR_data_prep.gdb => select New Feature Class => Call it “field_mapped_DoD” => set the type to “polygon” => On page 2 of the Create Feature Class menu, add a field to the attribute table called “change_elevation” => set the datatype as “Float”. Hit finish.

The image displays two sequential screenshots of the 'Create Feature Class' dialog box in ArcGIS.

Left Screenshot (Define Tab):

- Name:** field_mapped_DoD
- Alias:** (empty)
- Feature Class Type:** Polygon
- Geometric Properties:**
 - ☐ M Values - Coordinates include M values used to store route data.
 - ☒ Z Values - Coordinates include Z values used to store 3D data.
- ☒ Add output dataset to current map

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Buttons: Previous, Next, Finish, Cancel

Right Screenshot (Fields Tab):

- Fields Table:**

Field Name	Data Type
OBJECTID	OBJECTID (32-bit)
SHAPE	SHAPE
change_elevation	Float

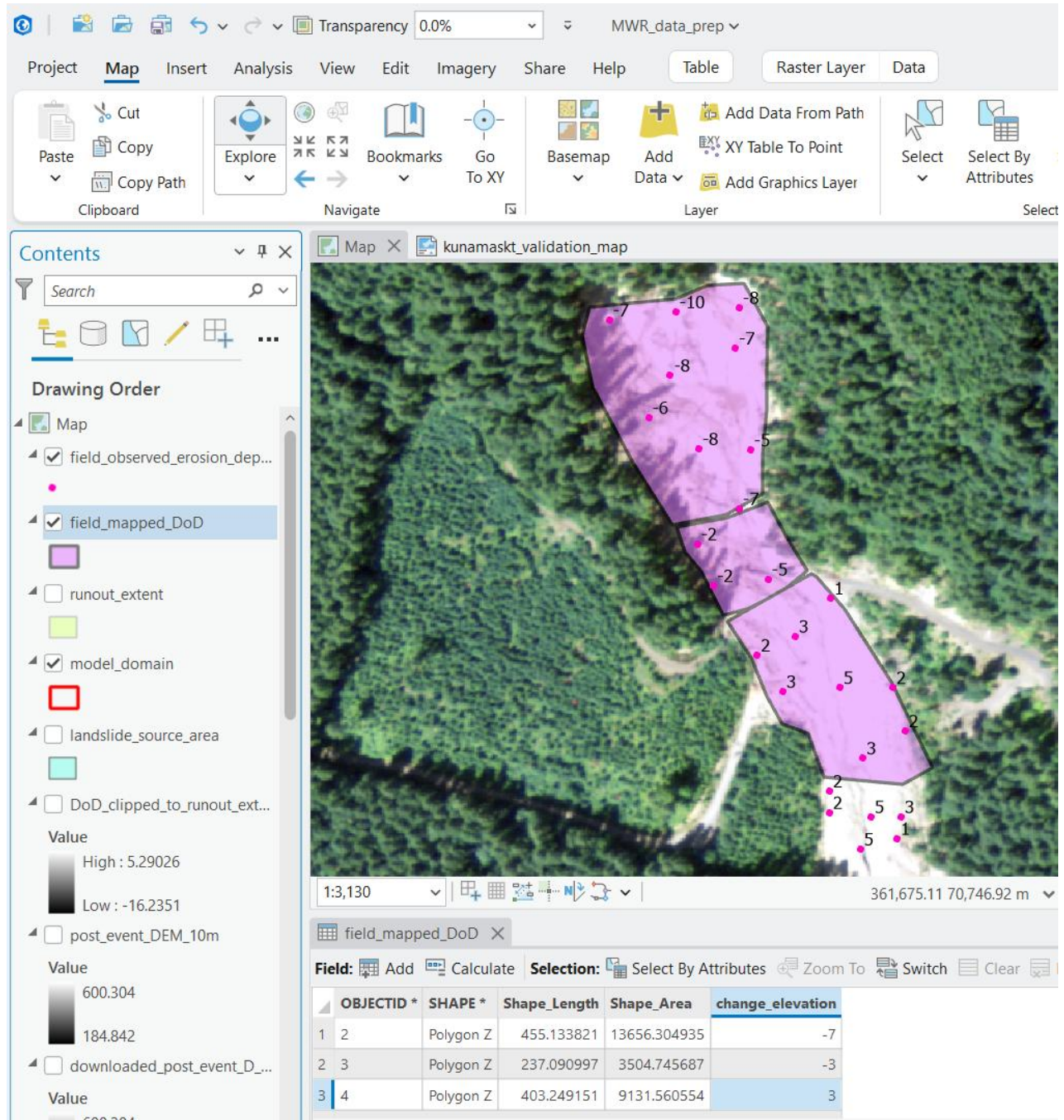
- Field Properties:**

Field Name	Properties
change_elevation	Alias: (empty), Allow Null Values: Yes, Default: (empty), Domain: (empty)

- Buttons: Import, Delete
- Page 2/6
- Buttons: Previous, Next, Finish, Cancel

At the bottom of the right screenshot, the 'Catalog' pane is visible, showing the 'Create Feature Class' dialog box as the active tool.

- g. In the Edit ribbon, hit the “Create” button. Select the “field_mapped_DoD” polygon you just created. Using your gps-located erosion/deposition depths as well as your field sketch, draw a polygon for each continuous area of relatively uniform elevation change. For example, I defined three areas of uniform elevation change in the upper part of the Cascades Mountains, 2009 runout path as shown below. While drawing each polygon, I also entered an average erosion/deposition value in the change_elevation column of the attribute table of the field_mapped_DoD polygon.



- h. Once you have finished defining areas of uniform erosion and deposition, go to the Edit ribbon and hit Save.

4. Convert the *field_mapped_DoD* polygon to a raster using the *Polygon to Raster* tool

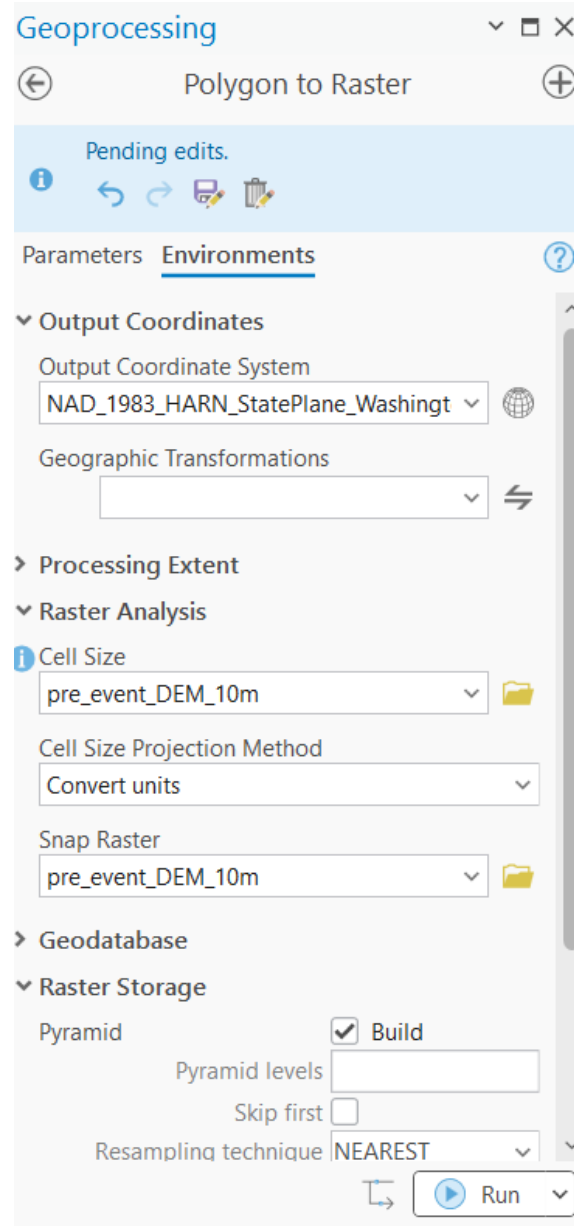
- a. For Input Features => select the *field_mapped_DoD* polygon. Set the Value field to *change_elevation* (or whatever column label you used for this) => set the save location and raster file name => enter a cell size to the same as the pre-event DEM (in this case 10). Don't hit run yet.

The screenshot shows the 'Polygon to Raster' tool in the QGIS Geoprocessing panel. The tool is titled 'Polygon to Raster' and has a 'Run' button at the bottom right. The 'Parameters' tab is selected, showing the following settings:

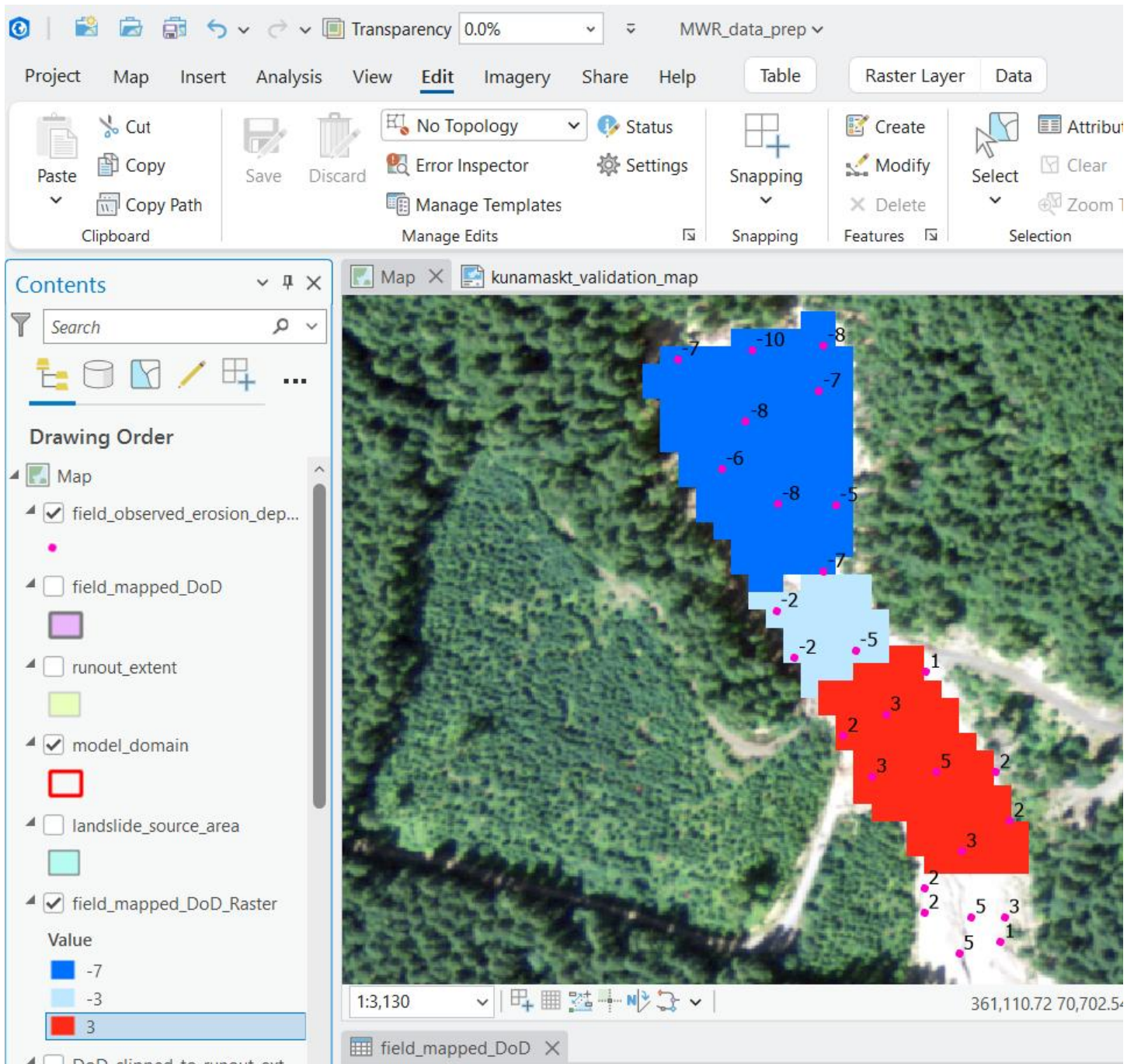
- Input Features:** field_mapped_DoD
- Value field:** change_elevation
- Output Raster Dataset:** field_mapped_DoD_raster
- Cell assignment type:** Cell center
- Priority field:** Shape_Area
- Cellsize:** 10
- Build raster attribute table:** ☒

The 'Cellsize' field is highlighted with a blue background. The 'Run' button is a blue circle with a white play icon and the text 'Run'.

- b. In the Environments Tab => set the Output Coordinate System, Processing Extent, Cell Size and Snap Raster to the pre-event DEM. Hit Run.



5. You should end up with a raster version of the field_mapped_DoD polygon something like below:



6. Finally, check that the grid of raster version of the field_mapped_DoD is the same as the pre_event_DEM_10m by right clicking on post_event_DEM_10m in the Contents pane and comparing the number of columns, rows and spatial reference.

Step 7. Define regolith (soil) thickness

In MWR, the grid cells that make of the initial landslide source area define where a landslide occurs but not its depth. Instead, landslide depth is defined by the regolith. The thickness of the regolith (all unconsolidated earth materials above the solid bedrock, including saprolite) defines both the thickness of the initial landslide and the amount of downslope erodible material.

1. *Gather soil thickness information*

a) Natural Resources Conservation Service:

- i) <https://websoilsurvey.nrcs.usda.gov/app/WebSoilSurvey.aspx>
- ii) <https://soilmap2-1.lawr.ucdavis.edu/gmap/>

b) Hike the watershed or terrain near the observed runout and measure soil thickness by inspecting road cuts and tree-throw pits.

2. *Define regolith thickness for the model domain*

a) As a function of contributing area and slope using a script by Westrick (1999):

https://github.com/pnnl/DHSVM-PNNL/blob/master/CreateStreamNetwork_PythonV/soildepthscript.py

b) Uniform thickness:

This is most easily done in Landlab with a line of code like:

```
grid.at_node["soil_thickness"] = np.ones(mg.number_of_nodes)*your_uniform_thickness
```

3. *Define regolith thickness at the landslide*

a. If regolith thickness at the landslide varies from the regional pattern, you can separately define regolith thickness at the grid cells defined by the initial_landslide_source_area raster shapefile. This is most easily done in Landlab with a line of code like:

i. For uniform thickness

```
grid.at_node["soil_thickness"][landslide_source_nodes] = your_uniform_landslide_thickness
```

ii. Thickness defined by the DoD

```
grid.at_node["soil_thickness"][landslide_source_nodes]=  
grid.at_node["DoD_observed"][landslide_source_nodes]*-1
```


Step 8. Convert layers to esrii ascii files

1. Convert shapefiles to an esrii ascii file using the *Raster to ASCII (Conversion Tools)* tool.

- For each raster (pre_event_DEM, DOD_clipped_to_runout_extent, landslide_source_area_raster) => select the file name. For Output ASCII raster file, set the directory and name of the file.



- After converting each raster shapefile to an esri ascii file, check the header of each to make sure the ncols, nrows, xllcorner, yllcorner and cellsize values match. All of these files will be loaded into Landlab onto the same model grid and must match. The NODATA_value can be different.

```
ncols      106
nrows      156
xllcorner  360802.34663041
yllcorner  69994.877819238
cellsize   10
NODATA_value -9999
```